

Extracellular matrices regulate extravasation journey of leukocytes and inflammatory tissue fate

Reviewed Preprint

v2 • November 21, 2025

Revised by authors

Reviewed Preprint

v1 • September 24, 2025

Yu-Tung Li 

Department of Stem Cell Therapy Science, Graduate School of Medicine, The University of Osaka, Suita, Japan

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This Review Article provides a timely review of how the extracellular matrix (ECM), particularly the vascular basement membrane, regulates leukocyte extravasation, migration, and downstream immune function, with a focus on monocytes/macrophages. It integrates molecular, mechanical, and spatial aspects of ECM biology in the context of inflammation, drawing from recent advances.

<https://doi.org/10.7554/eLife.108284.2.sa4>

Abstract

Leukocyte extravasation across the blood endothelium to inflamed tissues is a crucial defence mechanism against invading pathogens. After elimination of pathogen in the tissue, inflammation needs to be resolved back to steady-state. This cascade comprises of at least three stages: transmigration through the endothelium and the underlying basement membrane, intra-tissue leukocyte activity, and tissue resolution. In each stage, extracellular matrix proteins in the vascular basement membrane and in tissues regulate a multitude of endothelial and leukocyte functions essential to completion of the cascade, either as a collective force-permissive structure or through signalling by individual matrix proteins. Proper orchestration of these diverse processes during the extravasation journey ensures effective defence and avoids development of chronic inflammatory diseases. This review will focus on how these extracellular matrices regulate the extravasation journey of leukocytes to illustrate their tight functional inter-dependence with profound impacts on the ultimate post-inflammation tissue fate.

1. Introduction

Cells in an inflamed tissue secrete cytokines to activate local blood vessels and command leukocyte infiltration. Tissue entry of all kinds of leukocytes must be optimally controlled such that the cause of the inflammation, potentially an invading and readily proliferative pathogen, can be timely eliminated. Latency of tissue entry depends on the leukocyte types, typically with

neutrophils as the quickest responders (in hours), followed by monocytes (in hours to days) and lymphocytes (in days). Despite the different latency, they often follow common principles for tissue entry. Since the immune attack often involves cytotoxic substances such as granzymes and proteases, an overdose of leukocyte entry would cause an acute shock and tissue damages. Instead of a passive barrier for leukocytes to penetrate, the endothelium actively controls the leukocyte traffic via expression modulation of apical adhesion or surrogate molecules and junctional remodelling.^{1,2} Only specific sites of the blood vessels, with moderate to low shearing blood flow and optimal expression of supportive endothelial molecules, are permissive to leukocyte exit from the circulation (extravasation). These extravasation sites exist primarily in post-capillary venules, venules or specialised blood vessels such as the high endothelial venules (HEV) and the liver sinusoids.²⁻⁴ Understanding regulation details of these extravasation events is fundamental to devise treatments for various inflammatory diseases.

Along the extravasation journey to fight pathogens, leukocytes are prone to functional modulation, solely through interaction with their surroundings. The resultant leukocyte activities determine whether pathogens can be cleared, and if so, whether the inflamed tissue can be restored to homeostasis. Macrophages derived from either tissue residence or extravasated monocytes play a pivotal role in this latter resolution process.

Across this multi-stage paradigm, extracellular matrix (ECM) proteins in the vascular basement membrane and in tissue play significant regulatory roles. Overviews of the leukocyte extravasation processes, and intra-tissue leukocyte activities regulated by these ECM are given in **Figure 1**–**2** and **3** respectively. This review focuses on the regulatory roles of ECM along the extravasation journey of leukocytes to inflamed tissue. This journey has intimate involvement with endothelial cells (EC), especially with their junctional remodelling events. The processes regulated by ECM discussed in this review are summarised in **Table 1**. Following chronological order of the extravasation journey, we will describe what vascular structures constitute extravasation barriers for leukocytes (**Section 2**–**3**), how the leukocytes overcome these barriers (**Section 4**–**6**), how crossing these barriers affects leukocyte functions (**Section 7**), and how ECM interaction with leukocyte activities in tissue determines post-inflammatory tissue state (**Section 8**–**10**). At the end of this review, we will also discuss why control of macrophage cell state may hold the key for effective modulation of inflamed tissue state to revive tissue homeostasis and how different macrophage cell states regulate ECM functions (**Section 10**–**11**).

2. Venular architecture and barriers for leukocyte extravasation

In a blood venule, cells and extracellular components are organized in a defined manner to form extravasation barriers for leukocytes. The most inner circumference of the venule lines a monolayer of EC directly exposed to the shearing blood flow. Except for fenestrated blood vessels, this monolayer is typically continual, with individual EC intercalated via adherens junctions localized at the intercellular basolateral side of the plasma membrane. The endothelial adherens junctions contain various adhesion molecules, such as vascular endothelial cadherin (VE-cadherin), junctional adhesion molecules (JAMs), PECAM-1 and CD99.² In peripheral EC, tight junction molecules such as claudin-5 and ZO-1 are expressed in lower quantities, whereas strong expression of these molecules are detected in EC of some specialised tissues such as the blood-brain barrier and the blood-retinal barrier in brain and retina respectively. These junctional proteins are not rigidly held in place but under constant balancing actions of endocytosis and exocytosis of junctional proteins such that dynamic remodelling of the junctional tightness is possible. This in turn affects leukocyte traffic and vascular permeability. The most abluminal side

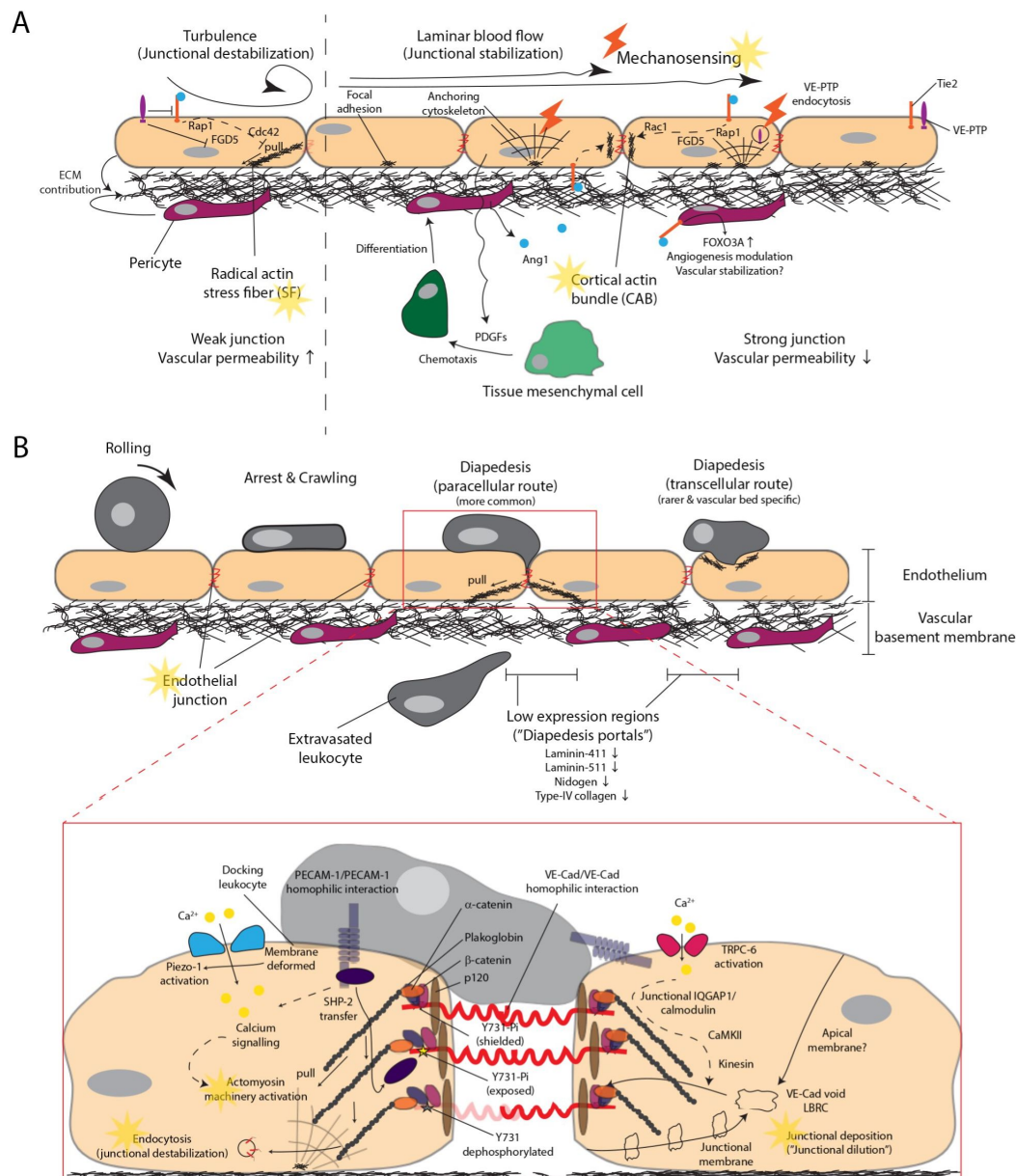


Figure 1

Vascular basement membrane enables endothelial mechanosensing and force-dependent junction remodelling to allow regulated leukocyte extravasation.

(A) Vascular basement membrane supports endothelial mechanosensing via focal adhesion to construct leukocyte extravasation barriers by regulating junctional strength and recruiting pericytes. Tie2 signalling is activated by laminar blood flow to trigger force-dependent actin filament remodelling to strengthen junctions. (B) After establishing firm endothelial interaction, leukocytes prefer to paracellularly transmigrate through the vascular junctional barrier near the low expression regions on the vascular basement membrane. (inset) After leukocyte signals diapedesis initiation via calcium signalling, the basement membrane allows force generation in endothelial cells to locally remodel VE-Cadherin junctional barrier for leukocyte passage. Gold spikes, processes regulated by the vascular basement membrane as a collective structure. Dashed line, multi-step signalling processes.

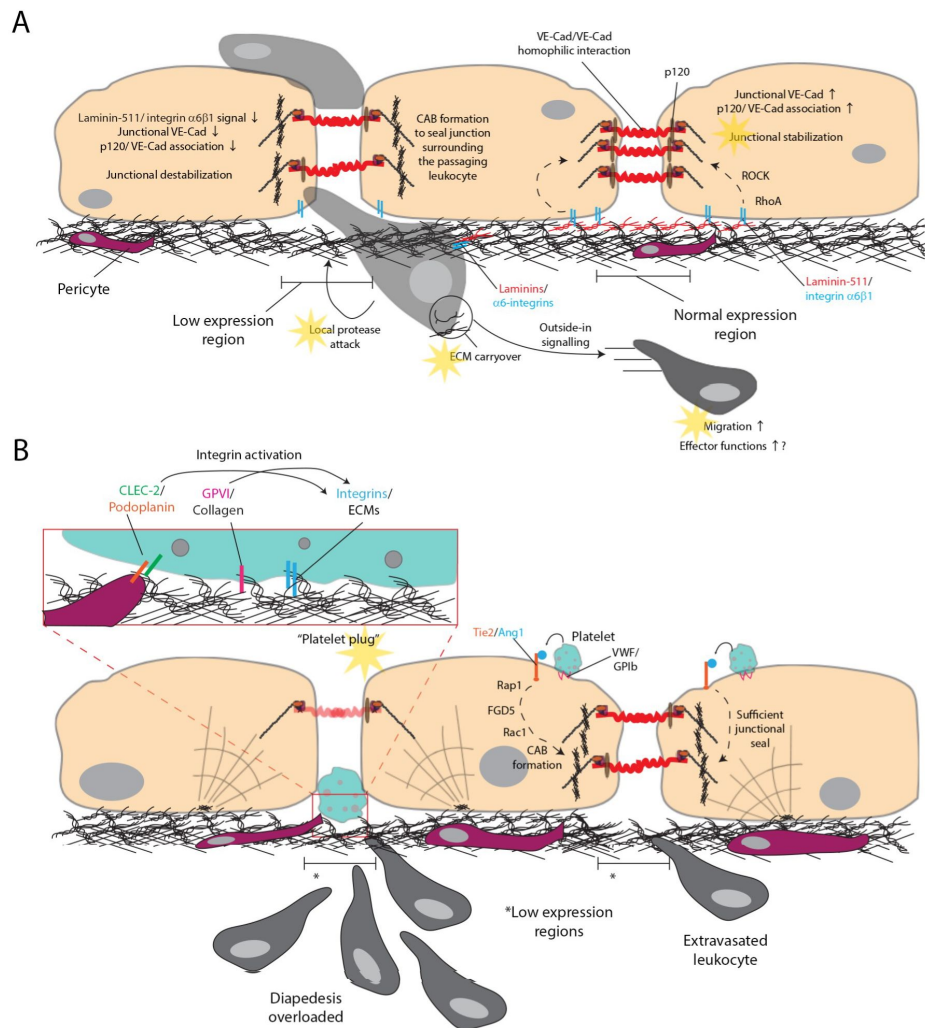


Figure 2

Regions in the vascular basement membrane with low expression of extracellular matrices are hotspots for leukocyte passage.

(A) Laminin-511 in the vascular basement membrane signals endothelial cells to stabilise junctions via remodelling VE-Cadherin trafficking and interaction with associated intracellular stabilising proteins. In the low expression regions, reduced local density of laminin-511, and thus the signalling to endothelial cells, loosens endothelial junction to ease leukocyte diapedesis. (B) Typically, platelet-endothelium Ang1-Tie2 interaction keeps junction surrounding the passing leukocyte tight to prevent leak. However, over-usage of low expression regions exposes the basement membrane for partial platelet activation generating a plug to prevent extravasation associated vascular leak. Gold spikes, processes regulated by specific ECM proteins from vascular basement membrane. Dashed line, multi-step signalling processes.

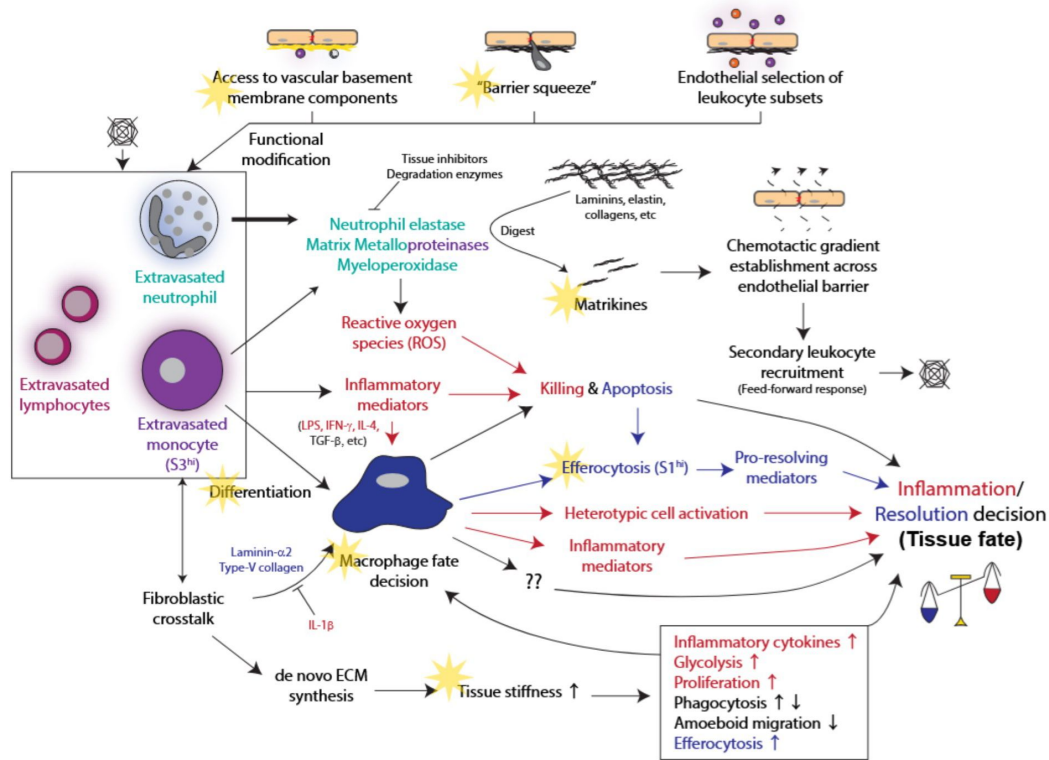


Figure 3

Extracellular matrices regulate various extravasated leukocyte activities in tissue and influence fate decision of the inflamed tissue.

Biophysical and biochemical cues experienced during vascular barrier passage modify functions of extravasated leukocytes in tissue. Leukocytes also gain access to regulatory ECM components on the abluminal side of the vascular basement membrane as well as in tissue. Enzymes produced by leukocytes cleave their respective target ECM components to generate chemotactic matrikines leading to secondary leukocyte extravasation. Besides directly executing inflammatory functions, cytokines produced by leukocytes regulate ECM production by tissue stromal cells, which modulate tissue stiffness and leukocyte functions via mechanosensing. These dynamic leukocyte-ECM interaction and activities integrate to the tissue fate decision between sustained inflammation and resolution. Gold spikes, processes regulated by ECM proteins from vascular basement membrane or in tissue.

Table 1

Biological processes regulated by extracellular matrix proteins or structures at each stage along the leukocyte extravasation journey.

Extravasation Journey Stage: Paracellular Trans-Endothelial Diapedesis		
ECM Interaction (Protein or Structure)	Function	Reference
Vascular Basement Membrane / Endothelial focal adhesions	Sense shear stress in blood flow to secrete PDGFs for pericyte recruitment and Ang1/Tie2-mediated junctional strengthening.	12,14
	Sense shear stress in blood flow to endocytose VE-PTP to activate Tie2-mediated junctional strengthening.	15
	Enable endothelial force generation pulling on VE-Cad/ catenins/ actomyosin complex to expose Y731 site for dephosphorylation by SHP-2, allowing subsequent VE-Cad endocytosis and local junctional destabilization around the passing leukocyte.	17,18
	Enable kinecin mediated delivery of LBRC along an anchored microtubule network to endothelial junction around the passing leukocyte for junctional destabilization.	15-17
Laminin-511 / Endothelial Integrin $\alpha 6 \beta 1$	Activate RhoA/ROCK to promote junctional deposition of VE-Cad to stabilize junction.	18
	Promote p120 association to stabilize junctional VE-Cad.	19
Extravasation Journey Stage: Trans-Basement Membrane Diapedesis		
ECM Interaction (Protein or Structure)	Function	Reference
Laminins / $\alpha 6$ Integrins	Allow migration through the basement membrane barrier.	41,44
	Locally breach the basement membrane barrier.	40,44,47
Vascular Basement Membrane / Leukocyte proteases	Produce ECM carryovers on extravasated leukocytes.	41,47
Laminin carryovers on neutrophils	(Unknown)	41,47
Lumican carryovers on neutrophils bound by Mac1	Promote neutrophil migration via inducing outside-in signalling.	45
Vascular Basement Membrane / Neutrophils	Deform passing neutrophils and activate Piezo1 to promote bactericidal activity in tissue.	44
Collagen-IV / Monocyte MMP9	Digest locally the collagen-IV barrier to support trans-basement membrane migration of T cells.	41
Extravasation Journey Stage: Immediately After Extravasation		
ECM Interaction (Protein or Structure)	Function	Reference
Collagen / Platelet GPIIb/IIIa	Partially activate platelets to adhere on and seal the exposed basement membrane at diapedesis sites with traffics overload.	16,51
Vascular Basement Membrane / Endothelial focal adhesions	Enable endothelial force generation stimulated by platelet Ang1-Tie2 activation to build junctional stabilizing cortical actin bundles surrounding a passing leukocyte.	15,18
Extravasation Journey Stage: In Tissue		
	Function	Reference
Laminin-511 / Monocyte Integrin $\alpha 6 \beta 1$	Promote monocyte differentiation to macrophage.	47
Collagen / MMP-8/9 and Prolyl Endopeptidase	Generate the matrilin PGP to secondarily chemoattract neutrophils, monocytes and T cells.	76,79,83
Laminin- $\gamma 2$ / Neutrophil Elastase	Generate the matrilin FGGPNCHEGAFSCPACYNQVKI to secondarily chemoattract neutrophils.	81
Elastin / Macrophage MMP-12	Generate the matrilin VGVAPG to secondarily chemoattract monocytes.	81,83
Laminin-511 / Leukocyte Proteases	Potentially generate fragments containing the matrilin sequence AQARSAASKVKVSMKF to secondarily chemoattract neutrophils and monocytes.	84
Versikine / ADAMTS1	Generate the matrilin (versikine) to modify macrophage cell state.	86
Versikine / TLR2 (and other unknown receptors)	Increase macrophage IL-1 β and IL-6 productions and decrease IL-10 production.	86
	Increase macrophage IL-10 production in the presence of immune complex.	86
	Stimulate tumour cell production of monocyte and T cell chemokines.	86,87
Stiff Tissue due to ECM Build-up / Leukocyte Mechanosensors	Increase inflammatory cytokine production.	94,95,97
	Shift leukocyte migration from amoeboid to podosomal mode.	96
	Modulate phagocytosis.	94,96,99
	Promote phagocyte proliferation.	97
	Promote phagocyte glycolysis.	97
$\alpha 2$ Laminins and Type-V collagen / (Unknown Monocyte Receptor)	Sensitize phagocyte activation response to stimuli.	97
	Promote efferocytosis by Piezo1 activation.	111
	Guide monocyte differentiation to efferocytic S1 macrophage by stimulating SHP-1 to suppress STAT-5 activity.	108

of the venule is decorated with scattered perivascular cells (pericytes). (**Figure 1A**) Lacking a definitive pan-pericyte marker, pericytes are defined as mesenchymal cells at vascular proximity, with subsets expressing markers of PDGFR β , CD146, NG2, and/or α -SMA.⁵

In between the EC monolayer and the porous pericyte layer is an ultra-thin adhesive layer of basement membrane composed of a network of self-assembled ECM proteins.^{6,7} Constituent ECM proteins mainly comprise of laminins (α 4 and α 5 chains), type-IV collagen, nidogens and perlecan.^{8,9} These components bind endothelial integrins organised in focal adhesions to allow EC adhesion,^{9,10} and support establishment of cell polarity and intercellular junctions to limit undesirable tissue access of cellular or soluble blood components. The EC cytoskeleton links to the focal adhesion-ECM structure to form a mechanosensory structure containing proteins (such as talin and vinculin) capable to translate mechanical force into biochemical signals.¹¹ This mechanosensing capacity, at least in part, organises pericytes to perivascular space and subcellular junctional architecture of EC. Laminar shear stress in the blood flow stimulates endothelial abluminal secretion of PDGF-B to recruit tissue mesenchymal cells and differentiate them to pericytes.^{12–14} These pericytes then release Ang1 to activate endothelial Tie2 signalling strengthening endothelial junctions. This junctional strengthening signal is further potentiated by endocytosis of the VE-PTP, which inhibits Tie2, by laminar shear stress.¹⁵ Since both EC and pericytes contribute to ECM components in the basement membrane, this pericyte recruitment process may help formation of nascent basement membrane in angiogenic vessels. (**Figure 1A**)

For a leukocyte to enter inflamed tissues, it engages in a series of steps well-described by the classical extravasation cascade.^{1,2,16} After firm adhesion on the endothelium, down-tuning of integrin activity allows the leukocyte to crawl on the endothelium and search for a suitable tissue entry site.^{17,18} At this stage, the continual endothelial monolayer and the underlying basement membrane are the sole barriers for leukocytes to overcome before tissue access. (**Figure 1B**) The porous abluminal pericyte layer, in contrast, does not barricade leukocyte access. These barrier crossing processes, known as diapedesis, need to be tightly controlled to permit extravasation without disrupting general vascular structural integrity.

3. Paracellular versus transcellular route of entry across the endothelial barrier

A leukocyte could transmigrate across the endothelium either by diapedesis through the junctions (paracellular route) or directly through the EC (transcellular route) in vitro.^{19–21} This observation questions whether the endothelial junction or the apical surface represents the major endothelial barrier for leukocytes under physiological conditions. If one route can be specifically blocked, assessment of its influence on the extravasation response shall inform the importance of the blocked route.

VE-cadherin is a major regulator of the adherens junction and supports junctional integrity by homophilic dimerization of VE-cadherin from adjacent cells. Like other cadherins, VE-cadherin is bound at the C-terminus by intracellular α -catenin and α -catenin, which is then connected to the actomyosin contraction machinery. (**Figure 1B**, inset) Since association to α -catenin stabilizes cadherin adhesion²², association or dissociation of catenin from VE-cadherin allows fine-tuning of junctional stability. In a previous study, transgenic mice with VE-cadherin genetically fused to α -catenin, thereby stabilizing the VE-cadherin junction and limiting paracellular diapedesis, were generated.²³ In these mice, the paracellular diapedesis events were reduced by ~50%, but the transcellular events were unaffected. In IL-1 α inflamed cremaster, LPS stimulated lung, and a skin model of delayed type hypersensitivity, both neutrophil and lymphocyte extravasation were

reduced by ~60%.²³ The similar reduction magnitudes of extravasation and paracellular diapedesis events suggest most of the reduced extravasation was due to blockade of paracellular diapedesis.

Although these observations suggest the endothelial junction is the sole barrier a leukocyte needs to overcome in vivo, “closing” the paracellular route did not affect lymphocyte entry to lymph nodes via the HEV, indicating vessel- or tissue-specific preference of diapedesis route.²³ Later, it was demonstrated that the density of ICAM-1, an adhesion molecule supporting leukocyte adhesion on the endothelium, presented by the EC could influence the route decision, that a high ICAM-1 density favours the transcellular route for helper T cells across brain EC.²⁴ Hence, the preference for transcellular or paracellular route appears to depend on the adhesion strength between the leukocyte and the endothelium and the vascular bed in concern. Nevertheless, in many inflammatory conditions, the paracellular route of transendothelial migration remains the major route.²³ The intercellular adherens junctions therefore are the major barriers for leukocyte to cross the endothelium. (Figure 1B)

4. Vascular basement membrane supports force-dependent endothelial junctional remodelling required for leukocyte diapedesis across the endothelium

Actomyosin contraction in EC is required to support leukocyte paracellular transmigration.^{25,26} The stable EC anchorage on the basement membrane via focal adhesions allows contractive force to be transmitted to actomyosin filaments at the cell periphery. It was hypothesized that such force pulling on endothelial junctional proteins helps open the junction for leukocyte passage, but molecular details had been obscure until recently.

Using a Förster resonance energy transfer (FRET) based molecular force sensor module inserted to the C-terminus of VE-cadherin, in which the FRET signal vanishes when the pulling force exceeds 4 pN, it has been elegantly demonstrated that endothelial mechanical force acting on VE-cadherin triggers a cascade of events that allow paracellular passage of a leukocyte.²⁷ Multiple tyrosine residues, which phosphorylation states control different endothelial functions, are present in VE-cadherin. Shown in vivo with knock-in point mutations, dephosphorylation of tyrosine at Y731 position is required for leukocyte extravasation to inflamed tissues.²⁸ Upon leukocyte docking, EC-leukocyte PECAM-1 homophilic interaction triggers calcium signalling to activate the actomyosin machinery that pulls on VE-cadherin. The pulling force exposes the catenin-shielded Y731 site for dephosphorylation by the phosphatase SHP2, and induces endocytosis of VE-cadherin to loosen the junction for leukocyte passage.²⁷ (Figure 1B, inset) Since the force magnitude a cell generates correlates to the substrate rigidity,²⁹ endothelial cells on a soft substrate cannot dephosphorylate Y731 showing force dependence of this process. Fluorescence lifetime imaging microscopy (FLIM) is a technique that measures the lifetime of a fluorophore, defined as the time the fluorophore stays excited, which depends on the fluorophore species and its surrounding microenvironment.³⁰ Alteration in the FRET signal, due to the pulled VE-cadherin tension sensor, incurs a shift in fluorescence lifetime detectable by FLIM. FLIM images showed that only the local junction surrounding the passing leukocyte was being pulled to not to compromise the overall junctional integrity.²⁷ Since not all junctions within the same EC were pulled, there involves a subcellular spatial restriction to confine the initiating calcium signal near the junction being pulled.

Such a subcellular cue might be provided by the leukocyte locally pressing the endothelial plasma membrane. A recent report describes the involvement of the tension sensitive cation channel Piezo-1 in transendothelial diapedesis. Leukocyte docking changes the tension in plasma membrane and provokes calcium influx via nearby Piezo-1 to activate the contractile actomyosin machinery.³¹ (Figure 1B, inset) This model implies localized calcium signal and thus the subsequent actomyosin contraction only occur close to the leukocyte dock site. One would still need to explain why the leukocyte arrest site is not always the diapedesis site that a leukocyte usually crawls over a distance before beginning diapedesis.^{18,32}

There are other endothelial force dependent processes supporting trans-endothelial diapedesis. Studies suggested directed membranous deposition, termed lateral border recycling compartments (LBRC), to the junction facilitates leukocyte passage. Deposition of LBRC to leukocyte-proximal junction requires force generation by the molecular motor kinesin along anchored microtubules.³³ Mechanistically, LBRC contains all other diapedesis-relevant junctional proteins but not VE-cadherin, and deposition to junctional site surrounding a transmigrating leukocyte dilutes the local VE-cadherin density and loosens the junction.³⁴ Regarding how LBRC spatially coordinates with the passing leukocyte so that deposition is only directed to the leukocyte-contacted junction, again calcium signalling is involved. With calcium sensor mice that fluorescently visualise calcium signalling, leukocyte diapedesis triggers calcium signalling localized to the proximal junction. In this model, EC-leukocyte PECAM-1 homophilic interaction initiates calcium influx through the nearby calcium channel TRPC6. This activates calmodulin, immobilized at the junction via IQGAP1 binding, and downstream CaMKII to drive junctional deposition of LBRC.³⁵ (Figure 1B, inset) TRPC6 can be activated by mechanical stress,³⁶ it might as well be activated by mechanical force exerted by leukocytes like it is the case for Piezo-1. While it is known that kinesin-microtubule machinery can be regulated by calcium signals,³⁷ the molecular details on its coupling to LBRC movement are unclear. Additional open questions like whether and why LBRC membrane is exclusively from junctional membrane, and how VE-cadherin is excluded from entering this compartment remain to be explored.

Overall, junction opening during paracellular transendothelial diapedesis involves numerous force dependent processes and the basement membrane anchors endothelial cytoskeleton to allow force generation in EC.

5. Pre-existing barrier weak spots with low laminin- α 5 expression guide leukocyte diapedesis

Organized laminin structures known as the “diapedesis portals” in the venular basement membrane directly signal endothelial junctional remodelling to regulate leukocyte diapedesis. Laminins are heterotrimers of the constituent chains (α -, β and γ - chains). Laminins α 4 β 1 γ 1 and α 5 β 1 γ 1 (laminin-411 and laminin-511) are the major laminins in the vascular basement membrane.⁸ In vitro assays showed that laminin-511, but not laminin-411, activates RhoA/ROCK in EC to trigger junctional translocation of VE-cadherin, which stabilizes the junction and suppresses leukocyte transendothelial migration.³⁸ In the presence of shearing blood flow, laminin-511 also promotes p120 association to VE-cadherin and further stabilizes its presence at the junction.³⁹ (Figure 2A) These junction stabilizing properties of laminin-511 lead to the postulation that, if there are sites on the basement membrane with lower laminin-511 density, endothelial junctions near those sites would be more permissive to leukocyte passage. Such sites have actually been demonstrated.

Careful and detailed examination of the venular basement membrane ultrastructure by confocal microscopy with 3D-reconstruction revealed that ECM component distribution is uneven and patchy with regions of low expression of laminin-511, laminin-411, nidogen and type-IV

collagen.^{40,41} The spatial analyses revealed the luminal sides of these low expression regions are close to endothelial junctions whereas the abluminal sides are typically not occupied by pericytes.⁴⁰ (Figure 1B) The reduced laminin-511 density at these regions locally “loosens” the proximal luminal endothelial junctions for easier leukocyte access. Indeed, leukocyte diapedesis events were observed close to these low expression regions.^{40,42} Trans-basement membrane diapedesis through these sites still requires $\alpha 6$ integrin to allow migration on laminins and protease activity to “drill through” the basement membrane.^{43,44} (Figure 2A) In mice lacking laminin- $\alpha 4$, there is a compensatory upregulation of laminin- $\alpha 5$ resulting in a less patchy expression in the basement membrane. These mice consequently largely lack laminin- $\alpha 5$ low expression regions and the resulting reduction in leukocyte extravasation confirms the facilitative role of the low expression regions.^{38,45}

The exact mechanism how these low expression regions are formed in the first place remains elusive. One possibility is that these low expression regions might be remnants of previous diapedesis events, which involve protease activities. As neutrophils pass through these low expression regions, these regions transiently enlarge with protease involvement including neutrophil elastase; the sizes recover to basal level over time.^{40,46} The enlargement is at least in part related to deformability of the leukocyte since the passage of the more rigid neutrophils, but not the more flexible monocytes, resulted in low expression region enlargement.⁴⁷ In parallel, gaps between pericytes on the abluminal side of the basement membrane enlarge.⁴⁸ An implication of these observation is that pericytes may have stretched these low expression regions in the basement membrane to enlarge them. Alternatively, the gap enlargement might be due to stretching by the rigid neutrophils or proteolytic activities followed by ECM redeposition. Direct visualization remains to be demonstrated, potentially by live confocal imaging and fluorescently labelled leukocytes, basement membrane and pericytes. Whether and which component(s) of the basement membrane could be labelled by fluorescent proteins without disrupting its native organization remain to be investigated. It is unclear whether de novo deposition or the postulated basement membrane stretching by pericytes contributes to the recovery. If de novo deposition is involved, it remains to investigate how the deposition is controlled as not to bury and destroy these sites.

6. How do leukocytes overcome barriers without vascular leak?

During paracellular transendothelial diapedesis, only the endothelial junction surrounding the passing leukocyte is loosened. When asked if such localized junctional destabilization alone is sufficient to avoid vascular leak, it seems not to be the case. Complementary mechanisms must be involved, or the diapedesed leukocyte would leave a confined but leaky pore behind. This diapedesis pore must be sealed, either by endothelial junctional remodelling immediately behind the passing leukocyte or by a simple plug to prevent leak. This essential seal is provided by platelets. The vascular basement membrane coordinates these platelet processes through transient exposure of collagens or providing a structural support enabling force-dependent endothelial junctional remodelling.

In thrombocytopenic mice, which lack circulating platelets, skin inflammation induced by either UVB, croton oil or immune complex (reverse passive Arthus reaction) caused bleeding at the inflamed site. This bleeding can be reduced by either leukocyte depletion or suppression of leukocyte-endothelial interaction via rolling/ adhesion blockade or chemokine neutralization. Importantly, bleeding spots colocalized with extravasation sites and specific blockade of leukocyte diapedesis with VE-Cadherin Y731F point mutation reduced bleeding. These suggest leukocyte diapedesis in the absence of platelets caused the bleeding.⁴⁹ Similar inflammatory bleeding could be observed in GPVI-/- mice which platelets are present in blood but do not express

GPVI.⁵⁰ GPVI is an ITAM-containing receptor that signals via downstream Syk and Btk and activates platelets upon binding collagen, which is present in the vascular basement membrane or in tissue. Transfusion of untreated, but not Syk or Btk inhibited, wildtype platelets to GPVI-/- mice rescued the inflammatory bleeding, indicating GPVI/Syk/Btk signalling is required.⁵⁰ These findings imply platelet contact with the collagenous vascular basement membrane signals and activates platelets via GPVI to plug the diapedesis pore left by the extravasated neutrophil. This has been recently directly visualized in a skin immune complex vasculitis model with real-time 4D intravital confocal microscopy that single platelets immediately adhered and plugged the blood vessel at the exact diapedesis site utilized by the passing neutrophil. Examination of ultra-structures with transmission electron microscopy found granules in the plugging platelets suggesting these platelets had not been fully activated and degranulated.⁵¹ Consistently, a previous report showed platelet granules are not required for bleeding prevention in this inflammation model.⁵² Simultaneous antibody blockade of both GPVI and CLEC-2 resulted in much more bleeding, as well as more platelet leakage to extravascular space beyond the basement membrane, than GPVI single blockade.⁵¹ CLEC-2 binds podoplanin that is present on pericytes but normally not on EC nor in the basement membrane. Since both GPVI and CLEC-2 are activating receptors and platelet adhesion on the underlying basement membrane ECM likely involves integrins, GPVI and CLEC-2 probably concert in transmitting inside-out signalling for integrin activation to support adhesive plugging. (**Figure 2B**)

There is another mechanism where platelets prevent vascular leak via endothelial junctional remodelling. Inflamed EC display von Willebrand factor (VWF) on the apical surface allowing platelet interaction.^{53,54} VWF binding via GPIb activates platelets to lumenally release angiopoietin-1 which activates Tie-2.⁵⁵⁻⁵⁷ Activated Tie-2 in EC phosphorylates downstream Rap1 and FGD5 which serves as a guanine exchange factor for Rac1 and Cdc42 to suppress stress fibres (SF) and to promote circumferential actin bundles (CAB).^{58,59} With a stable anchorage on the basement membrane such that tension of actin filaments could be exerted on junction, CAB strengthens whereas SF loosens endothelial junctions. (**Figure 2B**) Endothelial conditional knockout or siRNA knockdown of Tie-2 phenocopied neutrophil diapedesis induced bleeding observed in thrombocytopenic mice.⁵⁵ How this junction sealing process coordinates with leukocyte passage is unclear, or it just constantly seals any junctional pore incurred by leukocyte diapedesis.

If this Tie-2 centred mechanism is in place, why is platelet plugging necessary in immune complex-mediated inflammation to prevent diapedesis mediated bleeding? It was found that platelet plugging occurred not at random diapedesis site but was concentrated at hotspots experiencing on average 4 successive neutrophil diapedesis events.⁵¹ It thus appears that this Tie-2 mechanism is the default measure to seal the diapedesis pore, but may not catch up to the pace of leukocyte diapedesis in some inflammatory scenarios. On the other hand, leukocytes tend to take the partially breached, easier route for diapedesis. When the diapedesis breach overwhelms the Tie-2 mechanism to an extent that exposes the collagenous basement membrane, the platelet plug serves as the final defence line to prevent bleeding. (**Figure 2B**)

Both Tie-2 and the downstream FGD5 are substrates of the inhibitory phosphatase vascular endothelial protein tyrosine phosphatase (VE-PTP). The Tie-2 mechanism could likely be strengthened by suppressing VE-PTP activity to counter vascular destabilization in inflammatory conditions with too rapid leukocyte extravasation. This approach has been shown to be beneficial in other vascular pathologies (reviewed in ⁶⁰). Alternatively, there have been proposals to use platelets as a drug delivery vehicle to tumours exploiting the thrombotic leaky tumoral vasculature.^{61,62} As both luminal VWF binding and the platelet plug involve partial platelet activation, similar platelet engineering approaches may be attempted to deliver inflammatory modulators in inflammatory diseases.

After overcoming the endothelial and basement membrane barriers, leukocytes reach the inflamed tissue to perform a multitude of functions. Instead of one-off barrier-crossing events, recent researches propose that previous endothelial and ECM interactions might impose lasting modulatory effects on leukocyte functions in tissue. In tissue, leukocytes are exposed to plenty of ECM components that interactions further affect leukocyte behaviours in tissue. All these dynamic events integrate to the decision on the tissue inflammation status: to sustain or to resolve the inflammation. Proper decision is crucial to avoid failed defence responses or detrimental chronic inflammation. The following sections will describe and discuss these pivotal aspects influencing the inflammation fate.

7. Barrier passages modify leukocyte functions in tissue via biophysical stress, proteolysis, selective leukocyte passage and spatial access to functional regulators

When a leukocyte diapedes through the endothelial barrier, the cell body is heavily deformed with endothelial junctional proteins constantly tightening up the endothelial-leukocyte cell surface interface to prevent leakage. Similar deformation is recorded when a leukocyte passages through the basement membrane barrier with the uses of barrier-degrading proteases. These are biophysically and biochemically intense processes that might change the functional phenotype of the extravasated leukocyte. In addition, unique cellular interaction with EC or material interaction with the otherwise shielded basement membrane during extravasation may also modify leukocyte functions. Growing evidence have supported this view.

It has been recently shown that in neutrophils, “squeezing” through confined space, corresponding to either the endothelial or basement membrane barrier in vivo, activates the tension sensor Piezo-1 to trigger calcium signalling and to upregulate expression of the cytotoxic hydrogen peroxide producing enzyme NOX4⁶³, resulting in increased bactericidal activity.⁶⁴ Proteolytic activities involved during transbasement membrane diapedesis of neutrophils resulted in carryover of laminin bits on cell surface after arriving at tissues.^{40,47} While these bound laminins are bioactive as aforementioned, it is unclear whether they modulate neutrophil functions or interfere with crosstalk to other cell types. A potential hint comes from a similar carryover event with lumican, which is expressed by endothelial cells but not neutrophils. Lumican fragments were found on extravasated peritoneal neutrophils, but not on pre-extravasated blood or bone marrow neutrophils, and supported chemotactic migration and intra-tissue agility.⁶⁵ (Figure 2A) The enhanced motility likely involves lumican/Mac-1 mediated outside-in signalling. While outside-in signalling promotes leukocyte effector functions such as proliferation, cytokine production, myeloid degranulation and phagocytosis, it remains to be formally tested that whether leukocytes with lumican carryover show enhancement in these functions. Since laminin carryovers are also bound via integrins, modulation of outside-in signalling might represent a common mechanism of how ECM carryovers modify leukocyte functions. Another possibility is that lumican carryover might sensitise inflammatory responses as it bridges macrophage CD14 and bacterial LPS for better presentation to TLR4.⁶⁶

For monocytes, any functional influence by extravasation may occur immediately at the level of the monocyte, during macrophage differentiation, or at the level of differentiated macrophage. A previous study showed that following trans-basement membrane diapedesis, some extravasated monocytes remained associated with the basement membrane where laminin-511 promotes transition of Ly6C⁺MHC-II^{lo} monocytes to Ly6C⁺MHC-II^{hi} macrophages.⁶⁷ While this function is consistent with the perivascular macrophages abundantly seen, it is unclear whether this differentiation boost is possible because passing monocytes are in contact with the basement

membrane in the first place, or monocytes have a natural tendency to migrate towards blood vessels. The latter scenario would argue this basement membrane function is standalone and does not require the extravasation process, that basement membrane laminin-511 would support monocyte differentiation even if monocytes are artificially injected into tissue. Either way, physiologically, exposure to laminin-511 in the basement membrane is only possible after extravasation.

Besides, monocyte/macrophage functions are affected by ontogeny (embryonic progenitor derived versus bone marrow derived) and lineage origins.^{68,69} Previously described with inducible lineage tracing mice, the hematopoietic stem cells in bone marrow are partially derived from PDGFR α ⁺ mesenchymal progenitors transiently present during embryonic day E7.5-E8.5 (PDGFR α -lineage).⁷⁰ As a result, about one-fourth of circulating monocytes is from the PDGFR α -lineage. Interestingly, vasculatures in skin and colon selectively prefer extravasation of PDGFR α -lineage Ly6C⁺ monocytes. The responsible molecules nevertheless remain unclear. After differentiation to macrophages, PDGFR α -lineage cells show reduced DC-like characters and express a lower level of level of the inflammatory *Il1b* transcripts, suggesting less involvement in activating cellular engagement.⁷¹ Further mechanistic understanding of how PDGFR α -lineage macrophages are tilted to a less activating phenotype shall reveal new perspectives in inflammation control.

Collectively, functions of neutrophils and monocytes are modulated either directly via signalling events incurred by the extravasation processes, via selective extravasation of a leukocyte subset, or via novel access to function-modulating materials in tissues. (Figure 3)

8. Leukocyte proteases fragment tissue ECM to generate matrikines which promote secondary leukocyte trafficking

Proteases are powerful weapons utilized by innate leukocytes, especially neutrophils, to fight inflammation triggering invading pathogens after tissue entry. Extravasated neutrophils have experienced the “squeeze” by the endothelial and basement membrane barriers and hence exhibit even stronger aggression towards pathogens. Delivery of neutrophil proteases stored in intracellular granules could take several forms. Besides the typical degranulation, neutrophils may choose to release neutrophil extracellular traps (NET) which comprise of histone-bound chromatin, proteases, and additional offensive enzymes such as the reactive oxygen species (ROS) generating myeloperoxidase.⁷² While a neutrophil cannot survive chromatin loss, in exchange, the sticky NET efficiently entangles pathogens for concentrated protease attack. Alternatively, proteases may be released as exosomes where individual molecules are oriented to resist antiprotease actions and to achieve stronger activities than free proteases.^{73,74} However, protease activities not only destroy pathogens but also tissue ECM. Excessive protease activities could damage tissue integrity, as observed in chronic inflammation such as cystic fibrosis and chronic obstructive pulmonary disease.^{74,75}

Breakdown of tissue ECM produces a class of biologically active ECM degradation products termed matrikines, that regulate leukocyte trafficking. These matrikines are often small enough to permeate the endothelial barrier to form a functional chemotactic gradient in vivo. Interestingly, a matrikine tends to promote chemotaxis of the leukocyte type releasing its synthesizing protease(s), constituting a feedforward positive feedback loop. (Figure 3)

Proline-Glycine-Proline (PGP) is a tripeptide derived from collagen under actions of neutrophil protease MMP-8/9 and prolyl endopeptidase.⁷⁶ Acetylation at N-terminus of PGP enables chemotactic activity for neutrophils via binding chemokine receptor CXCR1/2.^{77–79} This

binding is explained by the structural similarity between PGP/CXCR2 and the binding of the cognate ligand KC to CXCR2.⁷⁹ Leukotriene A4 hydrolase (LTA4H), which generates the well-known strongly chemotactic leukotriene B4, serves a surprising function to degrade PGP.⁸⁰ Another peptide corresponding to the 597-618 region of human laminin-γ2 chain was found to be chemotactic for neutrophils. Release of this peptide requires neutrophil elastase activity.⁸¹ Proteolytic cleavage of elastin, abundantly found in elastic tissues such as skin and lung, by macrophage MMP-12 produces a hexapeptide VGVAPG to chemoattract monocytes via binding S-Gal receptor.^{82,83} Laminin-511 in the vascular basement membrane potentially produces matrikines. A synthetic peptide (AQARSAASKVKVSMKF) derived from laminin-511 chemoattracts neutrophils and monocytes in vitro.⁸⁴ But it is unknown if this fragment is physiologically generated and if bioactivity requires exact sequence and length.

Besides myeloid leukocytes, in vivo, matrikines in tissue appear to be effective to chemoattract T cells as well. Although this implies targeting matrikine regulation could simultaneously regulate T cell traffic, the effect might be indirect. For example, in a vasculitis model where human arteries were implanted and inflamed subcutaneously to immunodeficient mice, systemic administration of PGP increased extravasation of both T cells and monocytes. However, monocyte or macrophage activity of MMP9 facilitates T cell migration, presumably by cleaving collagen-IV in the vascular basement membrane during extravasation.⁸⁵ In human myeloma and colorectal cancer, tumours showing strong proteolytic activities on versican correlate to CD8⁺ T cell infiltration. Versican, solely produced by macrophages, is cleaved by stromal cell-derived protease ADAMTS1 to generate a N-terminal matrikine fragment termed versikine.^{86,87} Further in vitro analysis however suggested that versikine alters macrophage cell state which leads to production of T cell chemokines by tumour cells.⁸⁶ Interestingly, intact versican and versikine appear to have inverse effects on inflammation since the intact form promotes antiinflammatory IL-10 production while the fragment promotes inflammatory IL-1β production in macrophages.^{86,88}

Matrikines are regulated by intra-tissue availability of synthesis and degradation enzymes, as well as their regulators. For example, the presence of low-molecular weight hyaluronan stimulates tissue MMP-9 expression⁸⁹ and may promote formation of PGP. Smoking strengthens macrophage expression of MMP-12 and suppresses the degradative LTA4H activity in lung could respectively promote accumulation of VGVAPG and PGP.^{90,91} Bioactivity of MMPs, which are typically secreted as latent proproteins, requires activation. The zinc ion at the catalytic site, which is otherwise shielded by the thiol group of a cysteine residue of the pro-domain, needs to be exposed. Detailed activation mechanisms vary by the MMP species and are not fully known, but could involve cleavage of the pro-domain by furin or an active MMP or allosteric exposure of the catalytic site.⁹² On the other end, active MMPs could be inactivated by tissue inhibitors of metalloproteinases (TIMPs). Thus, matrikines contribute to secondary leukocyte recruitment caused by protease activities of extravasated leukocytes and further potentiate leukocyte activities in tissue. The multi-level regulation of matrikine ana-/catabolism implies their contribution to leukocyte activities in tissue likely vary by diseases; further investigations shall identify common features of conditions where managing matrikine-induced secondary leukocyte trafficking presents benefits against diseases.

9. Tissue stiffness provides biophysical cues to modify extravasated leukocyte activities

On the contrary, instead of ECM degradation, there are disease conditions like skin keloid, bleomycin induced pulmonary fibrosis and liver fibrosis (cirrhosis) where inflammatory activities of extravasated leukocytes promote amorphous collagenous deposition and harden the tissue. Keloid is considered as a form of pathological scarring where initial wounding does not properly resolve but persists as chronic inflammation with undesirable accumulation of ECM. Without

knowing the exact cause, multiple leukocyte types (including macrophages, mast cells and regulatory T cells), growth factors and cytokines that promote fibroblast proliferation (TGF- β , PDGFs, FGFs) and angiogenesis (FGFs, VEGFs), and additional pleiotropic cytokines (IL6, IGF-1) contribute to this condition.⁹³ Similarly, following an injury by excessive bleomycin, restless leukocyte activities in lung stimulate overgrowth of ECM-secreting fibroblasts. Cirrhosis could be caused by a variety of conditions, such as viral infection, prolonged alcohol overconsumption, bile flow obstruction, diets and metabolic malfunctions, which lead to chronic hepatitis and fibroblast activation generating scar tissues in liver. Further disease progression and functional decline could lead to liver failure and cancer. ECM build-up in these pathological scenarios alters tissue stiffness that translates to profound impacts on leukocyte functions. (Figure 3)

Many leukocytes are mechanosensitive and change their behaviours according to the surrounding stiffness. Analogous to the EC-basement membrane binding described earlier, leukocytes bind the surrounding ECM substrates via focal adhesion to transmit the mechano-signals. In general, tissue hardening due to ECM accumulation drives inflammatory functions of leukocytes. Monocytes and T cells enhance their secretion of inflammatory cytokines in a stiff environment.^{94,95} Most leukocytes migrate in amoeboid fashion in normal tissues. A study has shown that macrophages adopted this migration mode only on soft substrates (<88 kPa); on stiffer substrates (323 kPa), cells instead exhibited podosome-dependent migration.⁹⁶ Many other leukocyte qualities relevant to inflammation, such as proliferation, glycolytic metabolism and sensitivity to stimulus, are promoted by stiff substrates.⁹⁷ Phagocytic capacity may also be modulated by stiffness but contradicting reports exist.^{96,98} One reason for this is that stiffness regulates the expression of specific phagocytic receptors and could differently affect phagocytosis towards different targets. Macrophage mannose receptor (CD206), which binds and supports endocytosis of ovalbumin, was downregulated on a stiffer substrate (12-50 kPa), when compared to a softer substrate (2 kPa). Accordingly, ovalbumin phagocytosis was reduced on the stiffer substrate, but transferrin endocytosis was unaffected.⁹⁹ Interestingly, this is an effect triggered only by medium stiffness since cells cultured on hard plastic (3 GPa) expressed more CD206. Such a non-linear effect on phagocytosis and the different ranges of substrate stiffness studied may explain the discrepancy between reports. In parallel, non-linear cellular responses to substrate stiffness have also been observed in regulatory T cells. Under induction conditions for regulatory T cells, regulatory T cell transition from helper T cells was more effective on a softer (<860 kPa) substrate than a stiffer (2600 kPa) one.¹⁰⁰ However, in an even softer range of substrate stiffness (7.5-140 kPa), regulatory T cell formation was supported by the stiffer substrates via upregulating oxidative phosphorylation.¹⁰¹ Similarly, such non-linear stiffness responses have also been reported with helper and cytotoxic T cells.^{102,103} Of note, while stiffness responses by T cells require actomyosin cytoskeleton, partial inhibition of the actin polymerization, which slows remodelling or mechano-transduction, shifted the optimal stiffness. This optimal stiffness is also affected by the density of stimulatory ligands acting on T cells.¹⁰³ Empirically, tissue hardening often correlates to more severe inflammation. However, the non-linear stiffness responses described above imply that tissue stiffness might cooperate with many other response-modulatory factors to effect on leukocyte functions, and that its influence deserves to be examined in disease-specific manner. In vivo studies of stiffness effects (e.g. hydrogel implantation) however could be tricky without causing, or at least consideration of, structural disruption of the native tissue environment.

10. Macrophage efferocytosis regulated by ECM is critical to timely subside inflammation and restore tissue homeostasis

To avoid these pathological events caused by abnormal and prolonged leukocyte activities, inflammation must be timely tamed as soon as the triggering pathogen is cleared. While pathogen removal reduces inflammatory reaction, many of the reactions triggered such as the matrikine response and stiffness driven inflammation are self-sustaining. Active processes must be involved to stop inflammatory propagation.

Since neutrophils, which are the first to arrive at the inflamed tissue, have a relatively short lifespan (in days), a considerable number of apoptotic neutrophils has been accumulated by the time of pathogen clearance. Somatic cells damaged by inflammatory mediators add to the apoptotic counts. Macrophages are responsible for clearing up these dying cells to prevent necrotic release of intracellular materials that can trigger secondary tissue damages. Macrophages sense the chemotactic materials released by apoptotic cells and migrate towards them. Different from endocytosis of soluble and smaller particles, engulfment of apoptotic cells, either directly via receptor binding or indirectly via bridging adaptors, evokes intracellular signalling programs to acquire a pro-resolving cell state. This process, termed efferocytosis, produces anti-inflammatory mediators, such as IL-10 and resolvins, as end products.¹⁰⁴ (Figure 4) Efferocytosis thus serves as a switch for transiting inflammation to resolution. However, acquisition of the efferocytic phenotype is under competition of other fate guiding signals such as pathogen derived LPS, IFN- γ , TGF- β , IL-4 and IL-1 β , which promote killing responses, cross-activate other leukocytes like class-II helper T cells, or reduce tissue availability of efferocytic supportive signals.^{105–107} (Figure 3)

Efferocytosis could be facilitated with soluble factors produced by other cell types. Shown in ligation induced periodontitis and monosodium urate crystal induced peritonitis models, EC produces soluble DEL-1 that bridges phosphatidylserine exposed on apoptotic cells and integrin $\gamma 3$ on macrophages to ease recognition of target cells and facilitate efferocytic responses.^{104,108} Recently, fibroblastic ECM has been shown to guide monocyte differentiation to macrophage of strong efferocytic capacity in a vitamin D3 analogue induced atopic dermatitis model. In skin, monocytes mainly extravasate in hypodermis where laminin- $\alpha 2$ isoforms colocalize with type-V collagen. These ECM activate the phosphatase SHP1 to suppress STAT5 activity during CSF-1 driven differentiation to form FR β /CD163⁺ S1 macrophage, with high expression of both chemotactic and efferocytic receptors towards apoptotic cells.¹⁰⁹ Previous studies showed that macrophages respond to BMP signalling,^{110,111} and endogenous BMP signals may also be involved in forming S1 macrophages. Like STAT5, suppressing BMP signals directly induced the S1 phenotype and prevented further induction by ECM. Boosting S1 differentiation with exogenous laminin-211 resolved atopic dermatitis; failure to form sufficient S1 macrophage, on the other hand, induced eosinophil maladaptation tilting the dermatitis away from resolution.¹⁰⁹ Intriguingly, while tissue hardening due to ECM build-up generally supports inflammation, a recent study showed that it also enhances macrophage efferocytosis via Piezo-1 activation, serving as an apparent counter-measure for the stiffness-induced inflammatory responses.¹¹² (Figure 4) Efferocytosis effectors have been proven beneficial in various experimental infections in peritoneal cavity, liver, gastrointestinal tract, lung and even the brain.¹¹³ This process therefore presents an avenue for prospective clinical modulation of inflammatory diseases as reviewed in ¹¹⁴.

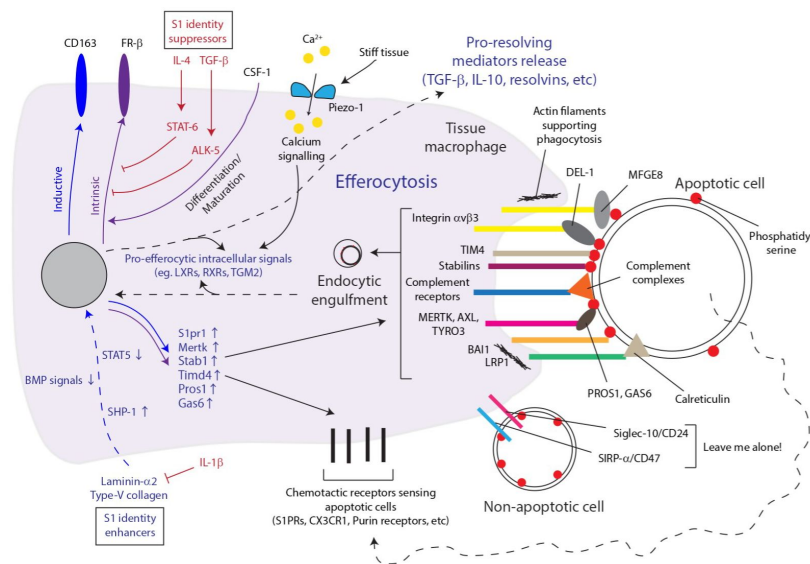


Figure 4

Efferocytosis exhibited by macrophages of strong S1 identity is an active driver of inflammation resolution and is regulated by specific extracellular matrices.

Macrophage senses chemotactic materials released by apoptotic cells via an array of receptors to migrate towards and engulf them. Engulfment is mediated by receptor recognition of phosphatidylserine exposed on apoptotic cells either directly or through sandwiching adaptors. Non-apoptotic cells avoid engulfment via specific receptor complexes. Intracellular processing of endocytosed apoptotic cells activates efferocytic programs and releases anti-inflammatory mediators. Efferocytic macrophage shows strong S1 identity and the efferocytic capacity is under regulation by external signalling crosstalk with cytokines and extracellular matrices.

However, efferocytic macrophages might not be protective in all scenarios. In kidney, CD206⁺ macrophages with transcript expression characteristics of the efferocytic S1 macrophages have been shown to cause renal fibrosis.¹¹⁵¹¹⁶ In line, the S1-promoting niche laminin- α 2 was found to accumulate in the fibrotic kidney in a genetic model of Alport disease.¹¹⁷ The hardening fibrotic renal tissue may further promote efferocytosis via Piezo-1 activation as well. Taken together, while both pro-inflammatory and pro-resolving mediators co-exist in the inflamed tissue, accumulative cell apoptosis acts as a timed switch to provoke efferocytosis which gradually tips the balance towards the resolving side. While efferocytosis modulation has immense therapeutic potential, caution must be taken in a pathology specific manner to evaluate its impacts on the disease.

11. Prospects in inflammatory tissue fate modulation via navigating macrophage cell states

The central role of macrophage efferocytosis in determining tissue inflammation fate and the beneficial effects presented by efferocytosis effectors in numerous pathologies.¹¹³ emphasize vast therapeutic potential of this type of macrophage and tempt medicine development to boost its formation. Although we showed that type-V collagen and α 2-laminins could serve this purpose,¹⁰⁹ their massive molecular structures hinder therapeutic uses. Rather, the downstream inactivation of STAT5 may be a more accessible target. Besides efferocytosis, this type of macrophage likely serves other functions, and whether these additional functions are beneficial, or at least unharmed, in all pathologies is debatable. It should be aware that pathological roles of macrophages have been frequently reported. Such dual nature of macrophage necessitates accurate control of macrophage cell state in pathology-specific manner. To effectively identify beneficial macrophage states with therapeutic potentials in various pathologies and to navigate other macrophage states to that direction, ideally there is a coherent descriptive platform for macrophage cell states, each annotated with guiding regulators. Under such a unified platform, macrophage states responsible for producing various aforementioned ECM regulators that tip the fate decision of tissue inflammation could be clarified. Nevertheless, this apparently simple task is particularly challenging for the extreme phenotypical diversity macrophages present in vivo.

Since the last century, flexibility in macrophage response has been demonstrated in vitro: IFN- γ triggers a classical activation response,¹⁰⁵ and IL-4 triggers an alternative response marked by mannose receptor upregulation.¹⁰⁶ In 2000, the concept of M1/M2 first emerged to describe different in vivo macrophage tendencies using arginine to produce either cytotoxic nitric oxide (M1) or ornithine (M2) in different mouse strains.¹⁰⁷ Later studies associated M1 with the classical response, inflammation and cytotoxicity, and M2 with the alternative response and anti-inflammation. Analysing more in vitro stimulus-specific macrophage responses found the M1/M2 dichotomy over-simplistic, and M2 response was subdivided into M2a, M2b, M2c and M4.¹¹⁸¹¹⁹ It has now been recognized that this conventional M1/M2 response scheme is insufficient to capture in vivo macrophage responses.¹²⁰ Transcriptome-wide analysis of tissue macrophages in bulk attributed the complex macrophage states in vivo to their adaptive responses to the diverse tissue microenvironment.¹²¹¹²² In an effort to comprehend the complex in vivo macrophage cell states with common principles shared by all tissues, we performed a meta-analysis on published single-cell macrophage transcriptomes from various pathophysiological tissues, and found macrophage populations shared between tissues in addition to reported tissue-specific macrophages. These shared macrophage populations could be segmented to five pan-tissue core identities (S1 to S5, defined by gene fingerprint). Under this framework, the macrophage transcriptomes and the identity kinetics during various pathologies were summarised in an archive termed Macrophage Identity Kinetics Archive (MIKA).¹⁰⁹ In order to cover more tissue types and conditions, we have expanded the archive to include 15 tissues with

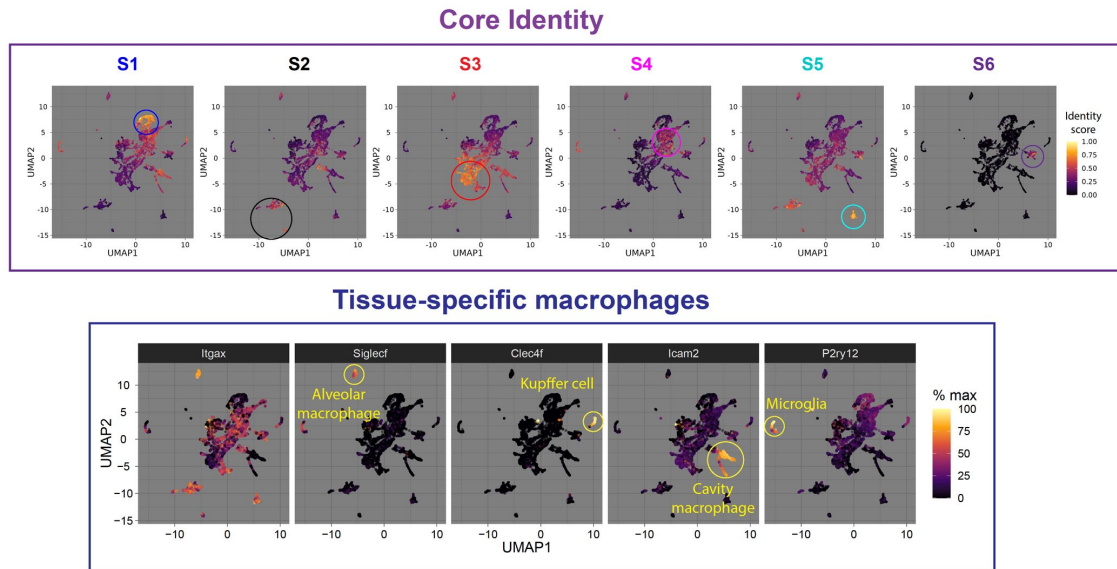
20 pathophysiologicals. During this expansion, we found additional pan-tissue cell states closely associated with the S1 identity. To address these S1-strays, we have replaced the original S4 identity, which has limited cell state defining efficiency, with two new core identities (new S4 and S6 identities). The current MIKA thus has six core identities in addition to tissue-specific macrophages. (**Figure 5A** [↗](#), Supplementary Data 1–4).

Under the MIKA framework, different macrophage identities could be examined for regulatory roles on ECM functional effectors described in this review. Macrophages contribute minimally to vascular matrices, with nidogen-2 very weakly expressed by S1 macrophages. Beyond this expression level and amongst the many collagen isoforms, type-XIV and type-XXVII exhibit prominent expression by S4 identity and microglia respectively, but functions of these rarer isoforms are less studied. Versican, which suppresses inflammation but generates the matrikine versikine upon proteolysis, is highly expressed by S3 identity. Instead of directly producing ECM, macrophages more often secrete fibroblastic growth factors, proteases and protease inhibitors to regulate ECM kinetics in tissue. Specific macrophage identities expressing these regulators are outlined below. S1 and S4 identities secrete different isoforms of PDGFs. While TGF- β 1 expression is detected in most identities, TGF- β 2 expression is limited to cavity macrophages. Matrikine-generating MMP-8, MMP-9 and MMP-12 are expressed by S3, S1 and S4 identities respectively. S1, S4 and several tissue-specific macrophages produce Serpin-B1 and -B6, which antagonise neutrophil proteases and may regulate matrikine production. Alveolar macrophages specifically express cathepsin K, which mediates phagocytic collagen catabolism and prevents lung fibrosis.¹²³ [↗](#) For other proteases or protease inhibitors with fibrosis modulating capacity (MMP-13, MMP-14, ADAM-19, ADAM-33, PAI-2 (Serpinb2), Serpin-B8, Serpin-E2 and TIMP-2), contribution by specific macrophage identities could also be observed. (**Figure 5B** [↗](#)) Overall, macrophages of S1, S3 and S4 identities as well as tissue-specific macrophages appear to be actively regulating ECM effectors with impacts on leukocyte functions. Of note, as S1 identity expresses PDGFs and MMP9, the strategy of promoting S1 macrophage formation to leverage efferocytosis for inflammation resolution shall be safe in conditions where these ECM regulators do not pose major concerns. In pathologies, macrophage identities dynamically fluctuate, so do their regulatory functions on the ECM functional effectors. Paying attention to these identity kinetics might hint ECM malfunction, which radiates to leukocyte traffic and functions, in conditions where the ECM-regulating macrophage identities expand. (Supplementary Data 2) Efforts to develop medicines that offer freedom to modulate macrophage identity could thus tune ECM functions, in addition to other macrophage functions, and influence the inflammatory tissue fate.

12. Concluding remarks

Extracellular matrix proteins are tightly coupled to the journey of a circulating leukocyte entering inflamed tissue. The vascular basement membrane anchors the endothelium to allow generation of cellular force required to remodel junctions for leukocyte passage. Leukocytes prefer using specialised diapedesis portals in the basement membrane, with weakened nearby endothelial junctions, to cross the vascular barriers. Optimal tissue entry requires aftercare of the “diapedesis pore” left behind by the extravasated leukocyte, that is fulfilled by cooperative efforts of collagens in basement membrane, platelets and EC. Both the vascular barrier passage and the myriad dynamic interaction with tissue cells and matrices influence leukocyte behaviours in tissue. These leukocyte activities, when properly orchestrated with macrophage fate, allow swift and timely restoration back to tissue homeostasis. This extravasation journey of leukocyte illustrates the intermingled relationship among extracellular matrices, leukocyte extravasation, tissue activities of extravasated leukocytes and the final tissue fate. We are still far from complete understanding of this extreme complexity. Yet, persistent stepwise advances in our understanding shall offer plenty regulatory targets for future modulation of inflammatory pathology.

A



B

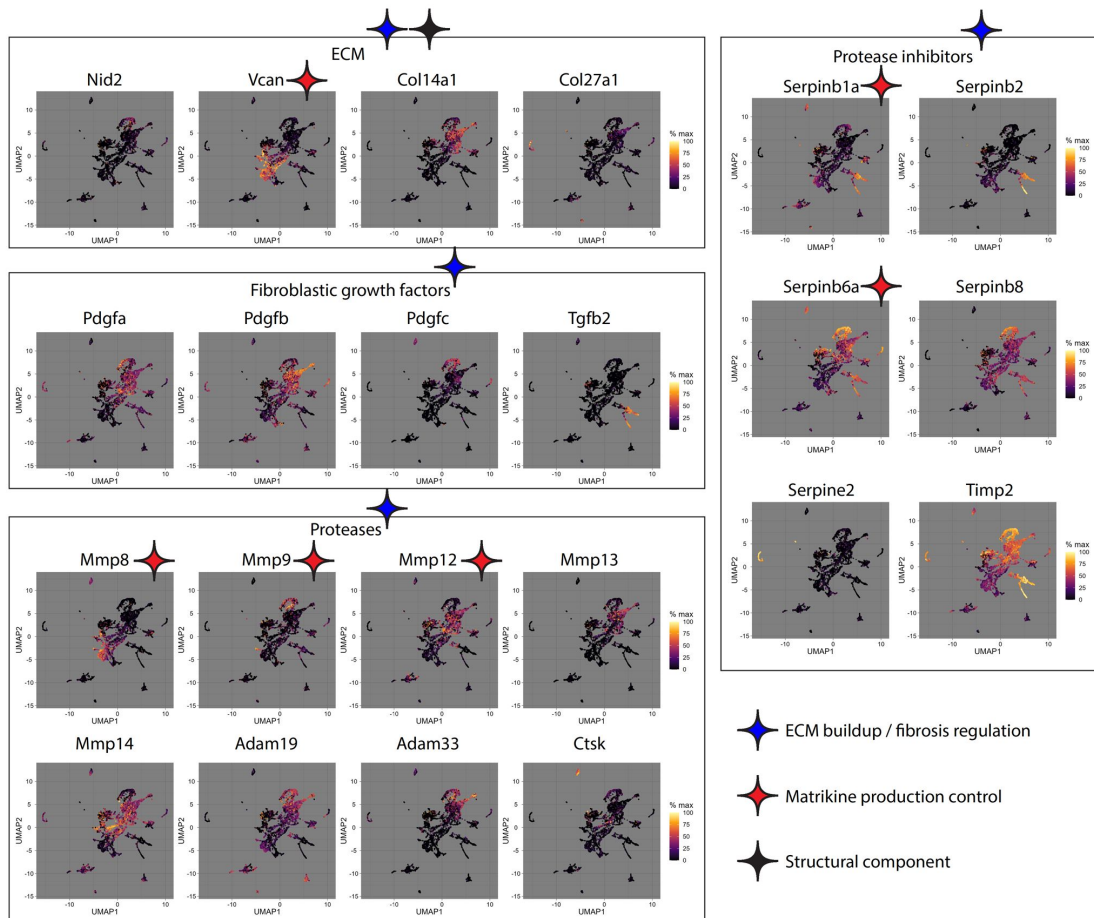


Figure 5.

Extracellular matrix functional effectors are regulated by specific identities of tissue macrophage.

(A) Under the pan-tissue macrophage identity framework MIKA, tissue macrophages are segmented to six core identities (S1-S6) and tissue-specific macrophages. (B) Macrophage identities expressing ECM structural components (black diamond), regulators of matrikine production (red diamond) or regulators of tissue ECM turnovers (blue diamond) are shown.

Appendix 1

Terms	Concise description
Diapedesis	The specific step(s) during leukocyte extravasation where a leukocyte transmigrates through a vascular barrier. May refer to trans-endothelial diapedesis or trans-basement membrane diapedesis. Different leukocyte and endothelial signalling events are involved depending on the barrier type and route (paracellular or transcellular) of diapedesis.
ECM carryover	After crossing the vascular basement membrane, fragments of the constituent ECM are bound on leukocytes as a result of proteolytic activities during the barrier passage. For some ECM, these fragments are reported to modulate leukocyte functions in tissue.
Efferocytosis	The process of macrophages (or phagocytes) engulfing apoptotic cells and thereafter synthesizing pro-resolving mediators. A process important to remove dead or dying cells accumulated throughout inflammation. Macrophages sense various cues released by apoptotic cells via a specific array of chemotactic receptors and migrate towards them. Surface exposure of phosphatidylserine on apoptotic cells triggers engulfment and subsequent pro-resolving signalling.
Extravasation	The entire multistep process of leukocyte crossing the vascular barriers and transmigrating from blood to tissue. Typically, a flowing leukocyte first tethers to the inflamed endothelium via catch-bond interaction with endothelial selectins. This initial tethering event is followed by decelerative rolling on endothelium and gradual activation of integrins on leukocytes. After firm arrest on the endothelium, the leukocyte crawls to search for a suitable site for trans-endothelial migration. Further trans-basement membrane migration and passage through any obstructing perivascular cells are required to gain access to the inflamed tissue.
Inflammatory tissue fate	An integrated decision that determines whether to resolve or to sustain an inflammation. This decision is dynamic and subjected to influence of myriad processes in the inflamed tissue, including the extravasation flux of leukocytes, leukocyte-ECM crosstalk during extravasation and in tissue, as well as other leukocyte and stromal cell activities. Correct decision at the right timing is crucial to promptly eliminate the initial

	cause of inflammation and to restore tissue homeostasis whilst preventing unnecessary tissue damages.
Matrikine	Bioactive peptide(s) produced from partial proteolysis of ECM. Different sets of proteases are responsible for digestion of different types of ECM. The resultant peptides are chemotactic to a specific leukocyte subset.
Mechanosensing	The process of a cell or cellular structure sensing the surrounding mechanical cues. This is achieved by binding to ECM present in the substrate via receptors (typically organized in focal adhesions) linked to the cytoskeleton decorated with mechanosensitive signalling proteins, which respond to and convert the mechanical cues to intracellular biochemical signals.
MIKA	A meta-transcriptome-based archive of tissue macrophage cell states and their kinetics in various pathophysiologies under a unified pan-tissue cell identity framework. Abbreviation of <u>M</u> acrophage <u>I</u> dentit <u>y</u> <u>K</u> inetics <u>A</u> rchive. The archive is expandable to accommodate new datasets. The current MIKA describes macrophages commonly found across multiple tissues and their kinetics with 6 core identities, S1 to S6, alongside with tissue-specific macrophages. The archive classifies complex macrophage states in tissue-borderless manner and serves as a common roadmap to navigate macrophage identity to pathology-resolving state(s) with identity regulators.
Secondary leukocyte extravasation	Additional leukocyte extravasation resulted from activities of extravasated leukocytes other than the initial cause(s) of the inflammation. This can be caused by de novo production of chemokines or matrikines (bioactive ECM fragments) from pre-existing ECM in tissue.
Tissue stiffness	The hardness of the tissue, or the elastic capability of the tissue to resist deformative force, usually expressed in Young's modulus. Tissue stiffness is affected by the amount, composition and alignment of ECM in tissue. While increase in tissue stiffness can be achieved by rapid ECM synthesis leading to tissue fibrosis, ECM usually have long turnover half-lives and stiffness decline or fibrosis resolution is a rather gradual process without active intervention of proteases. Through receptor binding and

	mechanosensing, tissue stiffness can modulate intra-tissue leukocyte functions.
Vascular barriers	In general, the endothelium and the vascular basement membrane are the sole barriers for leukocyte extravasation. Depending on the route of diapedesis (paracellular or transcellular), either the endothelial junction or the cell body constitutes the endothelial barrier.

Glossary of selected terms.

Acknowledgements

The author thanks Katsuto Tamai for discussion.

Additional information

Author contributions

Y.-T.L. wrote and edited the manuscript.

Sources of Funding

The work is supported by JSPS Kakenhi Grants-in-Aid for Scientific Research 25K19529.

Additional files

Supplementary data 1. Update documentation of MIKA. Description of minor modification of identity score computation, conditions and GEO accessions of newly integrated datasets, and addition of new identity elements auxiliary to S1 identity. (Flow chart) Expansion of the archive found S1-associated stray cell states (blue shade). The original S4 identity is replaced by two surrogate identities (new S4 and S6) to S1 identity to annotate the S1-strays (asterisks). Identity vector of each updated core identity is shown. Single hashtag indicates an ambiguous population with low gene expression. Tissue-specific states are marked by yellow circles or white arrows in tissue-expanded view. [↗](#)

Supplementary data 2. Macrophage identity scores of multiple tissue conditions in MIKA. Sample-wise macrophage identity proportions, tissue origin, perturbation, GEO accession and the definition of the input cell population for single-cell RNA sequencing were indicated. Identity proportions omitting monocyte-rich S3 were also provided, useful for datasets where baseline samples showed high S3 fraction. Note that the definition and experimental preparation of the input population affect subsequent efficiency of *in silico* purification of macrophages and the identity composition in a sample. This results in discrepancy of baseline identity between datasets of the same tissue (eg. GSE180420 and GSE200115). Identity kinetics is best assessed with the same definition of the input population (usually within the same dataset). [↗](#)

Supplementary data 3. Gene correlation to each macrophage identity in MIKA. For each tissue (baseline or perturbed), Pearson correlation coefficients of each gene to each of the macrophage identity (S1 to S6) are tabulated. Moran I coefficient shows how focused the gene expression is on UMAP. For searches of identity markers, the correlation should be considered in adjunct with gene expression (Supplementary data 4); a gene of high correlation to an identity but with low expression likely underrepresents the identity of interest. [↗](#)

Supplementary data 4. Gene expression of each elemental macrophage identity in MIKA. For each core macrophage identity (S1-S6) and tissue-specific macrophage (baseline or perturbed), average gene expression is tabulated in natural logarithmic scale. [↗](#)

References

1. Nourshargh S., Alon R. (2014) **Leukocyte migration into inflamed tissues** *Immunity* **41**:694–707 [Google Scholar](#)
2. Vestweber D. (2015) **How leukocytes cross the vascular endothelium** *Nat. Rev. Immunol* **15**:692–704 [Google Scholar](#)

3. Bogoslawski A., Butcher E. C., Kubes P. (2018) **Neutrophils recruited through high endothelial venules of the lymph nodes via PNAd intercept disseminating Staphylococcus aureus** *Proc. Natl. Acad. Sci. U. S. A* **115**:2449–2454 [Google Scholar](#)
4. McNamara H. A., Cockburn I. A. (2016) **The three Rs: Recruitment, Retention and Residence of leukocytes in the liver** *Clin. Transl. Immunol* **5**:e123 [Google Scholar](#)
5. Hirschi K. K., D'Amore P. A. (1996) **Pericytes in the microvasculature** *Cardiovasc. Res* **32**:687–698 [Google Scholar](#)
6. Lash J. M., Sherman W. M., Hamlin R. L. (1989) **Capillary Basement Membrane Thickness and Capillary Density in Sedentary and Trained Obese Zucker Rats** *Diabetes* **38**:854–60
7. Timpl R., Brown J. C. (1994) **The laminins** *Matrix Biol. J. Int. Soc. Matrix Biol* **14**:275–281 [Google Scholar](#)
8. Timpl R. (1996) **Macromolecular organization of basement membranes** *Curr. Opin. Cell Biol* **8**:618–624 [Google Scholar](#)
9. Pozzi A., Yurchenco P. D., Iozzo R. V. (2017) **The nature and biology of basement membranes** *Matrix Biol. J. Int. Soc. Matrix Biol* **57–58**:1–11 [Google Scholar](#)
10. Brown J. C., Sasaki T., Göhring W., Yamada Y., Timpl R. (1997) **The C-terminal domain V of perlecan promotes beta1 integrin-mediated cell adhesion, binds heparin, nidogen and fibulin-2 and can be modified by glycosaminoglycans** *Eur. J. Biochem* **250**:39–46 [Google Scholar](#)
11. Finney A. C., Stokes K. Y., Pattillo C. B., Orr A. W. (2017) **Integrin signaling in atherosclerosis** *Cell. Mol. Life Sci. CMLS* **74**:2263–2282 [Google Scholar](#)
12. Hsieh H. J., Li N. Q., Frangos J. A. (1991) **Shear stress increases endothelial platelet-derived growth factor mRNA levels** *Am. J. Physiol* **260**:H642–646 [Google Scholar](#)
13. Lindblom P., et al. (2003) **Endothelial PDGF-B retention is required for proper investment of pericytes in the microvessel wall** *Genes Dev* **17**:1835–1840 [Google Scholar](#)
14. Teichert M., et al. (2017) **Pericyte-expressed Tie2 controls angiogenesis and vessel maturation** *Nat. Commun* **8**:16106 [Google Scholar](#)
15. Shirakura K., et al. (2023) **Shear stress control of vascular leaks and atheromas through Tie2 activation by VE-PTP sequestration** *EMBO Mol. Med* **15**:e16128 [Google Scholar](#)
16. McEver R. P. (2015) **Selectins: initiators of leucocyte adhesion and signalling at the vascular wall** *Cardiovasc. Res* **107**:331–339 [Google Scholar](#)
17. Semmrich M., et al. (2005) **Importance of integrin LFA-1 deactivation for the generation of immune responses** *J. Exp. Med* **201**:1987–1998 [Google Scholar](#)
18. Li Y.-T., et al. (2019) **Blood flow guides sequential support of neutrophil arrest and diapedesis by PILR-α1 and PILR-a** *eLife* **8** [Google Scholar](#)
19. Carman C. V., Springer T. A. (2004) **A transmigratory cup in leukocyte diapedesis both through individual vascular endothelial cells and between them** *J. Cell Biol* **167**:377–388 [Google Scholar](#)

20. Ley K., Laudanna C., Cybulsky M. I., Nourshargh S. (2007) **Getting to the site of inflammation: the leukocyte adhesion cascade updated** *Nat. Rev. Immunol* **7**:678–689 [Google Scholar](#)
21. Nieminen M., et al. (2006) **Vimentin function in lymphocyte adhesion and transcellular migration** *Nat. Cell Biol* **8**:156–162 [Google Scholar](#)
22. Ozawa M., Kemler R. (1998) **Altered cell adhesion activity by pervanadate due to the dissociation of alpha-catenin from the E-cadherin.catenin complex** *J. Biol. Chem* **273**:6166–6170 [Google Scholar](#)
23. Schulte D., et al. (2011) **Stabilizing the VE-cadherin-catenin complex blocks leukocyte extravasation and vascular permeability** *EMBO J* **30**:4157–4170 [Google Scholar](#)
24. Abadier M., et al. (2015) **Cell surface levels of endothelial ICAM-1 influence the transcellular or paracellular T-cell diapedesis across the blood-brain barrier** *Eur. J. Immunol* **45**:1043–1058 [Google Scholar](#)
25. Garcia J. G., Verin A. D., Herenyiova M., English D. (1998) **Adherent neutrophils activate endothelial myosin light chain kinase: role in transendothelial migration** *J. Appl. Physiol. Bethesda Md 1985* **84**:1817–1821 [Google Scholar](#)
26. Saito H., Minamiya Y., Saito S., Ogawa J. (2002) **Endothelial Rho and Rho kinase regulate neutrophil migration via endothelial myosin light chain phosphorylation** *J. Leukoc. Biol* **72**:829–836 [Google Scholar](#)
27. Arif N., et al. (2021) **PECAM-1 supports leukocyte diapedesis by tension-dependent dephosphorylation of VE-cadherin** *EMBO J* **40**:e106113 [Google Scholar](#)
28. Wessel F., et al. (2014) **Leukocyte extravasation and vascular permeability are each controlled in vivo by different tyrosine residues of VE-cadherin** *Nat. Immunol* **15**:223–230 [Google Scholar](#)
29. Han S. J., Bielawski K. S., Ting L. H., Rodriguez M. L., Sniadecki N. J. (2012) **Decoupling substrate stiffness, spread area, and micropost density: a close spatial relationship between traction forces and focal adhesions** *Biophys. J* **103**:640–648 [Google Scholar](#)
30. Datta R., Heaster T. M., Sharick J. T., Gillette A. A., Skala M. C. (2020) **Fluorescence lifetime imaging microscopy: fundamentals and advances in instrumentation, analysis, and applications** *J. Biomed. Opt* **25**:071203 [Google Scholar](#)
31. Wang S., et al. (2022) **Mechanosensation by endothelial PIEZO1 is required for leukocyte diapedesis** *Blood* **140**:171–183 [Google Scholar](#)
32. Goswami D., et al. (2017) **Endothelial CD99 supports arrest of mouse neutrophils in venules and binds to neutrophil PILRs** *Blood* **129**:1811–1822 [Google Scholar](#)
33. Mamdouh Z., Kreitzer G. E., Muller W. A. (2008) **Leukocyte transmigration requires kinesin-mediated microtubule-dependent membrane trafficking from the lateral border recycling compartment** *J. Exp. Med* **205**:951–966 [Google Scholar](#)
34. Mamdouh Z., Chen X., Pierini L. M., Maxfield F. R., Muller W. A. (2003) **Targeted recycling of PECAM from endothelial surface-connected compartments during diapedesis** *Nature* **421**:748–753 [Google Scholar](#)

35. Dalal P. J., et al. (2020) **Spatiotemporal restriction of endothelial cell calcium signaling is required during leukocyte transmigration** *J. Exp. Med* **218**:e20192378 [Google Scholar](#)
36. Brayden J. E., Earley S., Nelson M. T., Reading S. (2008) **Transient receptor potential (TRP) channels, vascular tone and autoregulation of cerebral blood flow** *Clin. Exp. Pharmacol. Physiol* **35**:1116–1120 [Google Scholar](#)
37. Deavours B. E., Reddy A. S., Walker R. A. (1998) **Ca²⁺/calmodulin regulation of the Arabidopsis kinesin-like calmodulin-binding protein** *Cell Motil. Cytoskeleton* **40**:408–416 [Google Scholar](#)
38. Song J., et al. (2017) **Endothelial Basement Membrane Laminin 511 Contributes to Endothelial Junctional Tightness and Thereby Inhibits Leukocyte Transmigration** *Cell Rep* **18**:1256–1269 [Google Scholar](#)
39. Di Russo J., et al. (2017) **Endothelial basement membrane laminin 511 is essential for shear stress response** *EMBO J* **36**:183–201 [Google Scholar](#)
40. Wang S., et al. (2006) **Venular basement membranes contain specific matrix protein low expression regions that act as exit points for emigrating neutrophils** *J. Exp. Med* **203**:1519–1532 [Google Scholar](#)
41. Voisin M.-B., Probstl D., Nourshargh S. (2010) **Venular basement membranes ubiquitously express matrix protein low-expression regions: characterization in multiple tissues and remodeling during inflammation** *Am. J. Pathol* **176**:482–495 [Google Scholar](#)
42. Sixt M., et al. (2001) **Endothelial cell laminin isoforms, laminins 8 and 10, play decisive roles in T cell recruitment across the blood-brain barrier in experimental autoimmune encephalomyelitis** *J. Cell Biol* **153**:933–946 [Google Scholar](#)
43. Dangerfield J., Larbi K. Y., Huang M.-T., Dewar A., Nourshargh S. (2002) **PECAM-1 (CD31) homophilic interaction up-regulates alpha6beta1 on transmigrated neutrophils in vivo and plays a functional role in the ability of alpha6 integrins to mediate leukocyte migration through the perivascular basement membrane** *J. Exp. Med* **196**:1201–1211 [Google Scholar](#)
44. Wang S., Dangerfield J. P., Young R. E., Nourshargh S. (2005) **PECAM-1, alpha6 integrins and neutrophil elastase cooperate in mediating neutrophil transmigration** *J. Cell Sci* **118**:2067–2076 [Google Scholar](#)
45. Wu C., et al. (2009) **Endothelial basement membrane laminin alpha5 selectively inhibits T lymphocyte extravasation into the brain** *Nat. Med* **15**:519–527 [Google Scholar](#)
46. Voisin M.-B., et al. (2019) **Neutrophil elastase plays a non-redundant role in remodeling the venular basement membrane and neutrophil diapedesis post-ischemia/reperfusion injury** *J. Pathol* **248**:88102 [Google Scholar](#)
47. Voisin M.-B., Woodfin A., Nourshargh S. (2009) **Monocytes and neutrophils exhibit both distinct and common mechanisms in penetrating the vascular basement membrane in vivo** *Arterioscler. Thromb. Vasc. Biol* **29**:1193–1199 [Google Scholar](#)
48. Proebstl D., et al. (2012) **Pericytes support neutrophil subendothelial cell crawling and breaching of venular walls in vivo** *J. Exp. Med* **209**:1219–1234 [Google Scholar](#)

49. Hillgruber C., et al. (2015) **Blocking neutrophil diapedesis prevents hemorrhage during thrombocytopenia** *J. Exp. Med* **212**:1255–1266 [Google Scholar](#)
50. Gros A., et al. (2015) **Single platelets seal neutrophil-induced vascular breaches via GPVI during immune-complex-mediated inflammation in mice** *Blood* **126**:1017–1026 [Google Scholar](#)
51. Currie S. M., et al. (2022) **Confocal Real-Time Analysis of Cutaneous Platelet Recruitment during Immune Complex-Mediated Inflammation** *J. Invest. Dermatol* **142**:2724–2732.e3 [Google Scholar](#)
52. Deppermann C., et al. (2017) **Platelet secretion is crucial to prevent bleeding in the ischemic brain but not in the inflamed skin or lung in mice** *Blood* **129**:1702–1706 [Google Scholar](#)
53. Lowenstein C. J., Morrell C. N., Yamakuchi M. (2005) **Regulation of Weibel-Palade body exocytosis** *Trends Cardiovasc. Med* **15**:302–308 [Google Scholar](#)
54. Bernardo A., et al. (2005) **Platelets adhered to endothelial cell-bound ultra-large von Willebrand factor strings support leukocyte tethering and rolling under high shear stress** *J. Thromb. Haemost. JTH* **3**:562–570 [Google Scholar](#)
55. Braun L. J., et al. (2020) **Platelets docking to VWF prevent leaks during leukocyte extravasation by stimulating Tie-2** *Blood* **136**:627–639 [Google Scholar](#)
56. Chow T. W., Hellums J. D., Moake J. L., Kroll M. H. (1992) **Shear stress-induced von Willebrand factor binding to platelet glycoprotein Ib initiates calcium influx associated with aggregation** *Blood* **80**:113–120 [Google Scholar](#)
57. Bryckaert M., Rosa J.-P., Denis C. V., Lenting P. J. (2014) **Of von Willebrand factor and platelets** *Cell. Mol. Life Sci. CMLS* **72**:307–326 [Google Scholar](#)
58. Frye M., et al. (2015) **Interfering with VE-PTP stabilizes endothelial junctions in vivo via Tie-2 in the absence of VE-cadherin** *J. Exp. Med* **212**:2267–2287 [Google Scholar](#)
59. Braun L. J., et al. (2019) **VE-PTP inhibition stabilizes endothelial junctions by activating FGD5** *EMBO Rep* **20**:e47046 [Google Scholar](#)
60. Vestweber D. (2021) **Vascular Endothelial Protein Tyrosine Phosphatase Regulates Endothelial Function** *Physiol. Bethesda Md* **36**:84–93 [Google Scholar](#)
61. Nishikawa T., Tung L. Y., Kaneda Y. (2014) **Systemic administration of platelets incorporating inactivated Sendai virus eradicates melanoma in mice** *Mol. Ther. J. Am. Soc. Gene Ther* **22**:20462055 [Google Scholar](#)
62. Li Y.-T., Nishikawa T., Kaneda Y. (2016) **Platelet-cytokine Complex Suppresses Tumour Growth by Exploiting Intratumoural Thrombin-dependent Platelet Aggregation** *Sci. Rep* **6**:25077 [Google Scholar](#)
63. Nisimoto Y., Diebold B. A., Cosentino-Gomes D., Lambeth J. D. (2014) **Nox4: a hydrogen peroxidegenerating oxygen sensor** *Biochemistry* **53**:5111–5120 [Google Scholar](#)
64. Mukhopadhyay A., et al. (2024) **Trans-Endothelial Neutrophil Migration Activates Bactericidal Function via Piezo1 Mechanosensing** *Immunity* **57**:52–67.e10 [Google Scholar](#)

65. Lee S., Bowrin K., Hamad A. R., Chakravarti S. (2009) **Extracellular Matrix Lumican Deposited on the Surface of Neutrophils Promotes Migration by Binding to $\alpha 2$ Integrin*** *J. Biol. Chem* **284**:23662–23669 [Google Scholar](#)
66. Wu F., et al. (2007) **A novel role of the lumican core protein in bacterial lipopolysaccharide-induced innate immune response** *J. Biol. Chem* **282**:26409–26417 [Google Scholar](#)
67. Li L., et al. (2020) **Endothelial Basement Membrane Laminins as an Environmental Cue in Monocyte Differentiation to Macrophages** *Front. Immunol* **11**:584229 [Google Scholar](#)
68. Lavine K. J., et al. (2014) **Distinct macrophage lineages contribute to disparate patterns of cardiac recovery and remodeling in the neonatal and adult heart** *Proc. Natl. Acad. Sci. U. S. A* **111**:16029–16034 [Google Scholar](#)
69. Li F., et al. (2022) **Monocyte-derived alveolar macrophages autonomously determine severe outcome of respiratory viral infection** *Sci. Immunol* **7**:eabj5761 [Google Scholar](#)
70. Ding G., Tanaka Y., Hayashi M., Nishikawa S.-I., Kataoka H. (2013) **PDGF receptor alpha+ mesoderm contributes to endothelial and hematopoietic cells in mice** *Dev. Dyn. Off. Publ. Am. Assoc. Anat* **242**:254–268 [Google Scholar](#)
71. Li Y.-T., et al. (2022) **PDGFR α -lineage origin directs monocytes to trafficking proficiency to support peripheral immunity** *Eur. J. Immunol* **52**:204–221 [Google Scholar](#)
72. Brinkmann V., et al. (2004) **Neutrophil extracellular traps kill bacteria** *Science* **303**:1532–1535 [Google Scholar](#)
73. Vargas A., Roux-Dalvai F., Droit A., Lavoie J.-P. (2016) **Neutrophil-Derived Exosomes: A New Mechanism Contributing to Airway Smooth Muscle Remodeling** *Am. J. Respir. Cell Mol. Biol* **55**:450461 [Google Scholar](#)
74. Genschmer K. R., et al. (2019) **Activated PMN Exosomes: Pathogenic Entities Causing Matrix Destruction and Disease in the Lung** *Cell* **176**:113–126.e15 [Google Scholar](#)
75. Gaggar A., et al. (2008) **A Novel Proteolytic Cascade Generates an Extracellular Matrix-Derived Chemoattractant in Chronic Neutrophilic Inflammation** *J. Immunol. Baltim. Md 1950* **180**:5662–5669 [Google Scholar](#)
76. Gaggar A., et al. (2008) **A Novel Proteolytic Cascade Generates an Extracellular Matrix-Derived Chemoattractant in Chronic Neutrophilic Inflammation** *J. Immunol. Baltim. Md 1950* **180**:5662–5669 [Google Scholar](#)
77. Pfister R. R., Haddox J. L., Sommers C. I., Lam K. W. (1995) **Identification and synthesis of chemotactic tripeptides from alkali-degraded whole cornea. A study of N-acetyl-proline-glycine-proline and N-methyl-proline-glycine-proline** *Invest. Ophthalmol. Vis. Sci* **36**:1306–1316 [Google Scholar](#)
78. Pfister R. R., Haddox J. L., Sommers C. I. (1998) **Injection of chemoattractants into normal cornea: a model of inflammation after alkali injury** *Invest. Ophthalmol. Vis. Sci* **39**:1744–1750 [Google Scholar](#)
79. Weathington N. M., et al. (2006) **A novel peptide CXCR ligand derived from extracellular matrix degradation during airway inflammation** *Nat. Med* **12**:317–323 [Google Scholar](#)

80. Gaggar A., Weathington N. (2016) **Bioactive extracellular matrix fragments in lung health and disease** *J. Clin. Invest* **126**:3176–3184 [Google Scholar](#)
81. Mydel P., et al. (2008) **Neutrophil Elastase Cleaves Laminin-332 (Laminin-5) Generating Peptides That Are Chemotactic for Neutrophils** *J. Biol. Chem* **283**:9513–9522 [Google Scholar](#)
82. Senior R.M., et al. (1984) **Val-Gly-Val-Ala-Pro-Gly, a repeating peptide in elastin, is chemotactic for fibroblasts and monocytes** *J. Cell Biol* **99**:870–874 [Google Scholar](#)
83. Taddese S., Weiss A. S., Jahreis G., Neubert R. H. H., Schmelzer C. E. (2009) **H. In vitro degradation of human tropoelastin by MMP-12 and the generation of matrikines from domain 24** *Matrix Biol. J. Int. Soc. Matrix Biol* **28**:84–91 [Google Scholar](#)
84. Adair-Kirk T. L., et al. (2003) **A site on laminin alpha 5, AQARSAASKVKVSMKF, induces inflammatory cell production of matrix metalloproteinase-9 and chemotaxis** *J. Immunol. Baltim. Md 1950* **171**:398–406 [Google Scholar](#)
85. Watanabe R., et al. (2018) **MMP (Matrix Metalloprotease)-9-Producing Monocytes Enable T Cells to Invade the Vessel Wall and Cause Vasculitis** *Circ. Res* **123**:700–715 [Google Scholar](#)
86. Hope C., et al. (2016) **Immunoregulatory roles of versican proteolysis in the myeloma microenvironment** *Blood* **128**:680–685 [Google Scholar](#)
87. Hope C., et al. (2017) **Versican-Derived Matrikines Regulate Batf3-Dendritic Cell Differentiation and Promote T Cell Infiltration in Colorectal Cancer** *J. Immunol. Baltim. Md 1950* **199**:1933–1941 [Google Scholar](#)
88. Tang M., et al. (2015) **Toll-like Receptor 2 Activation Promotes Tumor Dendritic Cell Dysfunction by Regulating IL-6 and IL-10 Receptor Signaling** *Cell Rep* **13**:2851–2864 [Google Scholar](#)
89. Fieber C., et al. (2004) **Hyaluronan-oligosaccharide-induced transcription of metalloproteases** *J. Cell Sci* **117**:359–367 [Google Scholar](#)
90. Woodruff P. G., et al. (2005) **A Distinctive Alveolar Macrophage Activation State Induced by Cigarette Smoking** *Am. J. Respir. Crit. Care Med* **172**:1383–1392 [Google Scholar](#)
91. Noerager B. D., et al. (2015) **A Potential Role for Acrolein in Neutrophil-Mediated Chronic Inflammation** *Inflammation* **38**:2279–2287 [Google Scholar](#)
92. Springman E. B., Angleton E. L., Birkedal-Hansen H., Van Wart H. E. (1990) **Multiple modes of activation of latent human fibroblast collagenase: evidence for the role of a Cys73 active-site zinc complex in latency and a 'cysteine switch' mechanism for activation** *Proc. Natl. Acad. Sci. U. S. A* **87**:364–368 [Google Scholar](#)
93. Lee C.-C., et al. (2023) **An updated review of the immunological mechanisms of keloid scars** *Front. Immunol* **14**:1117630 [Google Scholar](#)
94. Saitakis M., et al. (2017) **Different TCR-induced T lymphocyte responses are potentiated by stiffness with variable sensitivity** *eLife* **6**:e23190 <https://doi.org/10.7554/eLife.23190> | [Google Scholar](#)
95. Fahy N., Menzel U., Alini M., Stoddart M. J. (2019) **Shear and Dynamic Compression Modulates the Inflammatory Phenotype of Human Monocytes in vitro** *Front. Immunol*

10:383 [Google Scholar](#)

96. Sridharan R., Cavanagh B., Cameron A. R., Kelly D. J., O'Brien F. J. (2019) **Material stiffness influences the polarization state, function and migration mode of macrophages** *Acta Biomater* **89**:47–59 [Google Scholar](#)
97. Chakraborty M., et al. (2021) **Mechanical Stiffness Controls Dendritic Cell Metabolism and Function** *Cell Rep* **34**:108609 [Google Scholar](#)
98. Adlerz K. M., Aranda-Espinoza H., Hayenga H. N. (2016) **Substrate elasticity regulates the behavior of human monocyte-derived macrophages** *Eur. Biophys. J* **45**:301–309 [Google Scholar](#)
99. Mennens S. F. B., et al. (2017) **Substrate stiffness influences phenotype and function of human antigenpresenting dendritic cells** *Sci. Rep* **7**:17511 [Google Scholar](#)
100. Shi L., Lim J. Y., Kam L. C. (2024) **Improving Regulatory T Cell Production Through Mechanosensing** *J. Biomed. Mater. Res. A* **112**:1138–1148 [Google Scholar](#)
101. Shi L., Lim J. Y., Kam L. C. (2023) **Substrate Stiffness Enhances Human Regulatory T Cell Induction and Metabolism** *Biomaterials* **292**:121928 [Google Scholar](#)
102. Hickey J. W., et al. (2019) **Engineering an Artificial T-Cell Stimulating Matrix for Immunotherapy** *Adv. Mater. Deerfield Beach Fla* **31**:e1807359 [Google Scholar](#)
103. Yuan D. J., Shi L., Kam L. C. (2021) **Biphasic response of T cell activation to substrate stiffness** *Biomaterials* **273**:120797 [Google Scholar](#)
104. Kourtzelis I., et al. (2019) **DEL-1 promotes macrophage efferocytosis and clearance of inflammation** *Nat. Immunol* **20**:40–49 [Google Scholar](#)
105. Nathan C. F., Murray H. W., Wiebe M. E., Rubin B. Y. (1983) **Identification of interferon-gamma as the lymphokine that activates human macrophage oxidative metabolism and antimicrobial activity** *J. Exp. Med* **158**:670–689 [Google Scholar](#)
106. Stein M., Keshav S., Harris N., Gordon S. (1992) **Interleukin 4 potently enhances murine macrophage mannose receptor activity: a marker of alternative immunologic macrophage activation** *J. Exp. Med* **176**:287–292 [Google Scholar](#)
107. Mills C. D., Kincaid K., Alt J. M., Heilman M. J., Hill A. M. (2000) **M-1/M-2 Macrophages and the Th1/Th2 Paradigm** *J. Immunol* **164**:6166–6173 [Google Scholar](#)
108. Choi E. Y., et al. (2008) **Del-1, an endogenous leukocyte-endothelial adhesion inhibitor, limits inflammatory cell recruitment** *Science* **322**:1101–1104 [Google Scholar](#)
109. Li Y.-T., Takaki E., Ouchi Y., Tamai K. (2024) **Guided monocyte fate to FRP/CD163+ S1 macrophage antagonises atopic dermatitis via fibroblastic matrices in mouse hypodermis** *Cell. Mol. Life Sci. CMLS* **82** [Google Scholar](#)
110. Zhao D., et al. (2022) **ALK1 signaling is required for the homeostasis of Kupffer cells and prevention of bacterial infection** *J. Clin. Invest* **132**:e150489 [Google Scholar](#)
111. Ihle C. L., et al. (2024) **Unique macrophage phenotypes activated by BMP signaling in breast cancer bone metastases** *JCI Insight* **9**:e168517 [Google Scholar](#)

112. Wang Y., et al. (2024) **Stiffness sensing via Piezo1 enhances macrophage efferocytosis and promotes the resolution of liver fibrosis** *Sci. Adv* **10**:eadj3289 [Google Scholar](#)
113. Dalli J. (2017) **Does promoting resolution instead of inhibiting inflammation represent the new paradigm in treating infections?** *Mol. Aspects Med* **58**:12–20 [Google Scholar](#)
114. Mehrotra P., Ravichandran K. S. (2022) **Drugging the efferocytosis process: concepts and opportunities** *Nat. Rev. Drug Discov* **21**:601–620 [Google Scholar](#)
115. Cheung M. D., et al. (2022) **Resident macrophage subpopulations occupy distinct microenvironments in the kidney** *JCI Insight* **7**:e161078 [Google Scholar](#)
116. Zimmerman K. A., et al. (2019) **Tissue-Resident Macrophages Promote Renal Cystic Disease** *J. Am. Soc. Nephrol. JASN* **30**:1841–1856 [Google Scholar](#)
117. Delimont D., et al. (2014) **Laminin α 2-mediated focal adhesion kinase activation triggers Alport glomerular pathogenesis** *PLoS One* **9**:e99083 [Google Scholar](#)
118. Mantovani A., et al. (2004) **The chemokine system in diverse forms of macrophage activation and polarization** *Trends Immunol* **25**:677–686 [Google Scholar](#)
119. Gleissner C. A., Shaked I., Little K. M., Ley K. (2010) **CXC chemokine ligand 4 induces a unique transcriptome in monocyte-derived macrophages** *J. Immunol. Baltim. Md 1950* **184**:4810–4818 [Google Scholar](#)
120. Orecchioni M., Ghosheh Y., Pramod A. B., Ley K. (2019) **Macrophage Polarization: Different Gene Signatures in M1(LPS+) vs. Classically and M2(LPS-) vs. Alternatively Activated Macrophages** *Front. Immunol* **10**:1084 [Google Scholar](#)
121. Suzuki T., et al. (2014) **Pulmonary macrophage transplantation therapy** *Nature* **514**:450–454 [Google Scholar](#)
122. Lavin Y., et al. (2014) **Tissue-resident macrophage enhancer landscapes are shaped by the local microenvironment** *Cell* **159**:1312–1326 [Google Scholar](#)
123. Fabrik I., et al. (2023) **Lung macrophages utilize unique cathepsin K-dependent phagosomal machinery to degrade intracellular collagen** *Life Sci Alliance* **6**:e202201535 [Google Scholar](#)

Author information

Yu-Tung Li

Department of Stem Cell Therapy Science, Graduate School of Medicine, The University of Osaka, Suita, Japan

For correspondence: tomli_yt@sts.med.osaka-u.ac.jp

Editors

Reviewing Editor

Frederic Bard

Centre de Recherche en Cancérologie de Marseille, Marseille, France

Senior Editor

Tadatsugu Taniguchi

The University of Tokyo, Tokyo, Japan

Reviewer #1 (Public review):

Summary:

In this review, the author covered several aspects of the inflammation response, mainly focusing on the mechanisms controlling leukocyte extravasation and inflammation resolution.

Strengths:

This review is based on an impressive number of sources, trying to comprehensively present a very broad and complex topic. The revised version strengthens the connection with the ECM and all sections are now better integrated.

<https://doi.org/10.7554/eLife.108284.2.sa3>

Reviewer #2 (Public review):

Summary:

The manuscript is a timely and comprehensive review of how the extracellular matrix (ECM), particularly the vascular basement membrane, regulates leukocyte extravasation, migration, and downstream immune function. It integrates molecular, mechanical, and spatial aspects of ECM biology in the context of inflammation, drawing from recent advances. The framing of ECM as an active instructor of immune cell fate is a conceptual strength.

Strengths:

- Comprehensive synthesis of ECM functions across leukocyte extravasation and post-transmigration activity.
- Incorporation of recent high-impact findings alongside classical literature.
- Conceptually novel framing of ECM as an active regulator of immune function.
- Effective integration of molecular, mechanical, and spatial perspectives.

Weaknesses:

- Some sections remain dense with signalling detail.
- Figure readability could be improved through simplified labeling.

Appraisal and Impact:

The authors have achieved their aim of presenting an integrated view of ECM-immune interactions. The review provides conceptual and visual clarity on a complex topic.

<https://doi.org/10.7554/eLife.108284.2.sa2>

Reviewer #3 (Public review):

Summary & Strengths:

This review by Yu-Tung Li sheds new light on the processes involved in leukocyte extravasation, with a focus on the inter between leukocytes and the extracellular matrix. In doing so, it presents a fresh perspective on the topic of leukocyte extravasation, which has been extensively covered in numerous excellent reviews. Notably, the role of the extracellular matrix in leukocyte extravasation has received relatively little attention until recently. This review synthesizes the substantial knowledge accumulated over the past two decades in a novel and compelling manner.

The author discusses the relevant barriers leukocytes face during extravasation, addresses interactions with and transmigration through endothelial junctions, mechanisms supporting extravasation, and how minimal plasma leakage is achieved during this process. The question whether extravasation affects leukocyte differentiation and properties is original and thought-provoking and has received limited consideration thus far. The consequences leukocytes extracellular matrix interaction, non-linear responses to substrate stiffness and effects on macrophage polarization, efferocytosis and the outcome of inflammation are relevant topics raised. Finally, a unifying descriptive framework MIKA is introduced, which provides a tool for classifying macrophages based on their expression patterns and could inform the development of targeted therapies aimed at modulating macrophage identity and improving outcomes in inflammatory scenarios.

In summary, this review provides a stimulating perspective on leukocyte extravasation in the context of extracellular matrix biology.

Weaknesses:

One potential drawback of this review is that the attempt to integrate a vast amount of information has resulted in complex figures, which may lead to important details being overlooked by readers.

<https://doi.org/10.7554/eLife.108284.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

In this review, the author covered several aspects of the inflammation response, mainly focusing on the mechanisms controlling leukocyte extravasation and inflammation resolution.

Strengths:

This review is based on an impressive number of sources, trying to comprehensively present a very broad and complex topic.

Weaknesses:

(1) This reviewer feels that, despite the title, this review is quite broad and not centred on the role of the extracellular matrix.

Since this review focuses on the whole extravasation journey of leukocyte, this topic is definitely quite broad and covers several related fields. The article highlights the involvement of extracellular matrices (ECM), which are important regulators in multiple phases of the process, as a common theme to thread together these related topics. In the revised manuscript, we have made further emphasis on the role of specific ECM where appropriate (see point 2 below) and reorganized the last section to fit to this theme (see point 3 below).

(2) The review will benefit from a stronger focus on the specific roles of matrix components and dynamics, with more informative subheadings.

ECM may exert their roles either as a collective structure or as individual components. In the latter case, though the concerned ECM are specifically named throughout the manuscript, they may not be sufficiently obvious since they were often not mentioned in subheadings. For sections discussing functions of a specific ECM protein or at least a specific class of ECM proteins, we have now included their names in the subheadings as well for clarity (section 5 and 8). For other sections discussing functions that involve ECM as a macrostructure, either in form of vascular basement membrane to enable force generation or contributing to the overall tissue stiffness to provide biophysical cues (section 7, 9-10), we have included the specific processes regulated in the subheadings like that in section 4.

In the newly added discussion about the effects of matrikines on lymphocytes, we have also focused on the roles of specific ECM (PGP and versican; line 396-408). We hope these measures have made the subheadings more informative and provided better clarity of the roles of specific ECM components.

(3) The macrophage phenotype section doesn't seem well integrated with the rest of the review (and is not linked to the ECM).

Section 10-11 concerns how macrophage phenotypes affect the tissue fate following inflammation, that is, either to resolve inflammation and regenerate damages incurred or to sustain inflammation. This fate decision is an important aspect of this review: By furthering our understanding on the processes and mechanisms involved, we hope to gain the capability to properly control tissue outcomes in inflammatory diseases.

In section 10, an emphasis is put on macrophage efferocytosis, for its documented efficiency to resolve tissue inflammation. Specific ECM components (type-V collagens and α 2-laminins) could directly promote macrophage efferocytosis (line 494-499). On the other hand, changes in tissue stiffness, as a result of ECM turnover regulated by activities of leukocytes or other cell types like fibroblasts as described in section 9, also affects efferocytosis (line 504-507).

We acknowledge that section 11 does not integrate well to the rest of the review, this section is now restructured. First, we describe how the ECM-regulated efferocytosis may be leveraged in disease modulation (line 522-529) and the need for a unified system to describe macrophage states for disease modulation (line 527-533) such that the responsible cell states for producing ECM regulators / effectors can be clarified (line 533-535). Given means to control macrophage cell states, this clarification will be useful to modulate pathologies involving ECM malfunctioning, that might be hinted by emergence or expansion of those responsible macrophage states in pathology (line 577-579, 581-585). Next, we provide historic background of efforts to establish such a unified descriptive platform for macrophage states (line 538-548) and describe the recent solution offered by MIKA. MIKA is a pan-tissue archive for tissue macrophage cell states based on meta-analysis of published single-macrophage transcriptomes, we have described the establishment, the latest development (Supplementary Data 1-4) and how the complex tissue macrophage states are segmented to core and tissue-specific identities under this framework (line 548-560, Figure 5A). Under this identity

framework, expression of different ECM regulators discussed in this review (either the ECM per se, fibroblastic growth factors or proteases or protease inhibitors that regulate ECM turnover or matrikine production) are examined and linked to specific macrophage identities to offer insights of their potential relevance in pathologies (line 561-586, Figure 5B).

(4) Table 1 is difficult to follow. It could be reformatted to facilitate reading and understanding

We apologize for the complex setup. Table 1 is now reformatted to horizontal orientation to have enough space for the columns and reorganized for much easier comprehension.

(5) Figure 2 appears very complex and broad.

The original Figure 2 is now split to 2 separate figures (Figure 3-4). Since many processes of diverse natures influence tissue decision of resolution/inflammation, Figure 3 serves to outline and summarise these processes. Figure 4 now focuses on the regulation and tissue-resolving roles of macrophage efferocytosis, which specific ECM components (type-V collagens and α 2-laminins) or tissue stiffness contribute to acquisition of this cell state. We hope this split can better focus the messages and ease understanding.

(6) Spelling and grammar should be thoroughly checked to improve the readability.

The manuscript is now proofread again, with corrections made throughout the text.

Reviewer #2 (Public review):

Summary:

The manuscript is a timely and comprehensive review of how the extracellular matrix (ECM), particularly the vascular basement membrane, regulates leukocyte extravasation, migration, and downstream immune function. It integrates molecular, mechanical, and spatial aspects of ECM biology in the context of inflammation, drawing from recent advances. The framing of ECM as an active instructor of immune cell fate is a conceptual strength.

Strengths:

- (1) Comprehensive synthesis of ECM functions across leukocyte extravasation and post-transmigration activity.*
- (2) Incorporation of recent high-impact findings alongside classical literature.*
- (3) Conceptually novel framing of ECM as an active regulator of immune function.*
- (4) Effective integration of molecular, mechanical, and spatial perspectives.*

Weaknesses:

- (1) Insufficient narrative linkage between the vascular phase (Sections 2-6) and the in-tissue phase (Sections 7-10).*

A transition paragraph between these two phases is now added between Section 6 and Section 7 to provide a narrative that ECM interaction events during extravasation affect downstream leukocyte functions (line 300-307).

- (2) Underrepresentation of lymphocyte biology despite mention in early sections.*

Although lymphocytes follow a similar extravasation principle as described in earlier sections, their in-tissue activities differ much from innate leukocytes. Discussion of crosstalk amongst T cells, innate leukocytes and matrikines is now incorporated into section 8 (line 396-408). Functional effects of tissue stiffness on different T cell subsets are now discussed in section 9 (line 456-469).

| (3) *The MIKA macrophage identity framework is only loosely tied to ECM mechanisms.*

The involved section 11 is now restructured to better integrate to the ECM topics with the associated Figure 3 changed to Figure 5. Specifically, under the MIKA framework, we have now linked specific macrophage identities to expression / production of ECM functional effectors or regulators discussed in this review to highlight their regulatory roles and potential relevance in pathologies. Reviewer #1 and #3 also have raised this issue, please refer to the response to point (3) of reviewer #1 for detailed description.

| (4) *Limited discussion of translational implications and therapeutic strategies.*

Besides translational implications or therapeutic strategies included in the original manuscript (line 291-298, 375-377, 421-424, 427-429, 508-511, 512-516 of the current manuscript), we have now included additional discussion to enrich these aspects (line 356-358, line 396-398, 402-403, 428, 436-439, 467-469, 523-536, 579-586).

| (5) *Overly dense figure insets and underdeveloped links between ECM carryover and downstream immune phenotypes.*

The original Figure 1 containing the insets is now split to Figure 1-2 to avoid too dense information fitting to a single figure and to better focus the message in each figure. To resolve the issue of overly dense insets, insets in Figure 1 are redrawn/ reorganized. The original Figure 1C is moved to Figure 2A. The inset showing platelet plugging, together with the issue of diapedesis overloading described in the original Figure 1B, is reorganized to Figure 2B. In this way, Figure 1 focuses on the vascular barrier organization, overview of extravasation, and the force related events during endothelial junctional remodelling. Figure 2 focuses on the low expression regions, and junctional sealing processes after diapedesis.

We have now expanded discussion on ECM carryovers and their reported or implicated effects on downstream leukocyte functions (line 329-335).

| (6) *Acronyms and some mechanistic details may limit accessibility for a broader readership.*

A glossary explaining specialized terms that may be confusing to readers of different fields is now included as Appendix 1 to broaden accessibility (line 977).

Reviewer #3 (Public review):

Summary & Strengths:

This review by Yu-Tung Li sheds new light on the processes involved in leukocyte extravasation, with a focus on the interaction between leukocytes and the extracellular matrix. In doing so, it presents a fresh perspective on the topic of leukocyte extravasation, which has been extensively covered in numerous excellent reviews. Notably, the role of the extracellular matrix in leukocyte extravasation has received relatively little attention until recently, with a few exceptions, such as a study focusing on the central nervous system (J Inflamm 21, 53 (2024) doi.org/10.1186/s12950-024-00426-

6) and another on transmigration hotspots (*J Cell Sci* (2025) 138 (11): jcs263862 doi.org/10.1242/jcs.263862). This review synthesizes the substantial knowledge accumulated over the past two decades in a novel and compelling manner.

The author dedicates two sections to discussing the relevant barriers, namely, endothelial cell-cell junctions and the basement membrane. The following three paragraphs address how leukocytes interact with and transmigrate through endothelial junctions, the mechanisms supporting extravasation, and how minimal plasma leakage is achieved during this process. The subsequent question of whether the extravasation process affects leukocyte differentiation and properties is original and thought-provoking, having received limited consideration thus far. The consequences of the interaction between leukocytes and the extracellular matrix, particularly regarding efferocytosis, macrophage polarization, and the outcome of inflammation, are explored in the subsequent three chapters. The review concludes by examining tissue-specific states of macrophage identity.

Weaknesses:

*Firstly, the first ten sections provide a comprehensive overview of the topic, presenting logical and well-formulated arguments that are easily accessible to a general audience. In stark contrast, the final section (Chapter 11) fails to connect coherently with the preceding review and is nearly incomprehensible without prior knowledge of the author's recent publication in *Cell. Mol. Life Sci. CMLS* 772 82, 14 (2024). This chapter requires significantly more background information for the general reader, including an introduction to the Macrophage Identity Kinetics Archive (MIKA), which is not even introduced in this review, its basis (meta-analysis of published scRNA-seq data), its significance (identification of major populations), and the reasons behind the revision of the proposed macrophage states and their further development.*

The issue of section 11 being not well-integrated to the rest of the review has also been pointed out by other reviewers. In response, this section and the associated Figure 3 are now restructured for better integration to the theme of ECM. In brief, we have now discussed the regulatory roles of specific macrophage identities under the MIKA framework on the ECM regulators described in this review. Please refer to the response to point (3) of reviewer #1 for further details.

Regarding the difficulties in understanding the MIKA framework without prior knowledge of our previous work, first, we thank the reviewer for pointing out this issue and for making suggestion to better introduce the framework in a way easy to comprehend. Accordingly, in the current structure of section 11, we have described the rationales behind the needs of a common descriptive platform for tissue macrophage states (line 523-536), previous historic efforts (line 538-548), have introduced MIKA with mentions of the establishment and significance (line 548-555), and also have explained the rationales behind further development (line 555-560).

Secondly, while the attempt to integrate a vast amount of information into fewer figures is commendable, it results in figures that resemble a complex puzzle. The author may consider increasing the number of figures and providing additional, larger "zoom-in" panels, particularly for the topics of clot formation at transmigration hotspots and the interaction between ECM/ECM fragments and integrins. Specifically, the color coding (purple for leukocyte $\alpha 6$ -integrins, blue for interacting laminins, also blue for EC $\alpha 6$ integrins, and red for interacting 5-1-1 laminins) is confusing, and the structures are small and difficult to recognize.

We apologize for the figures being too dense. Other reviewers have also raised this issue (see response to point (5) of reviewer #2 and response to point (5) of reviewer #1). The original Figure 1 and 2 are now reorganized to Figure 1-2 and 3-4 respectively, with insets also redrawn / expanded. Figure 1 now focuses on the vascular barrier organization, overview of extravasation, and the force related events during endothelial junctional remodelling. Figure 2 focuses on the low expression regions, and junctional sealing processes after diapedesis. Figure 3 serves to outline and summarise the diverse processes influencing tissue decision of resolution/inflammation. Figure 4 focuses on the regulation and tissue-resolving roles of macrophage efferocytosis. The original Figure 3, mainly concerning the methodological aspects of update of MIKA, is now integrated to Supplementary Data 1. This figure is now replaced as Figure 5 concerning the specific macrophage identities producing ECM effectors / regulators discussed in this review.

The concerned colour-coding issue is now in Figure 2A. All integrins are now in sky blue and all laminins in red. VE-Cad is also in red but has a different size and shape than laminins. We hope these modifications have improved the figures avoiding confusion.

Recommendations for the authors:

As you will see, the reviewers thought your manuscript was interesting and timely. However, as part 11 and its corresponding Figure 3 seem somewhat detached from the rest of the manuscript, one recommendation would be to remove this part for improved clarity. Other recommendations can be found in the comments below.

Reviewer #2 (Recommendations for the authors):

(1) Improve narrative linkage between vascular extravasation (Sections 2-6) and in-tissue leukocyte activities (Sections 7-10) by adding explicit transition text that connects ECM changes during transmigration to downstream immune cell phenotypes.

A transition paragraph is now added between section 6 and 7 (line 300-307).

(2) Expand discussion of lymphocyte-ECM interactions, either within existing sections or as a dedicated subsection.

We have now added discussion of the effects of matrikine on in vivo T cell traffic (line 396-409) and how T cell functions are regulated by tissue stiffness (line 457-466).

(3) Strengthen integration of the MIKA macrophage identity framework with ECM-specific drivers (e.g., stiffness, matrikines) and reduce methodological detail in Fig. 3 to focus on biological relevance.

We thank the reviewer for this recommendation and have adopted accordingly. First, the methodological details in the original Fig.3 is now integrated to Supplementary Data 1. This figure is now replaced as Fig.5 serving to examine different macrophage identities' contribution to ECM effectors / regulators (specifically, ECM per se, growth factors for ECM-producing fibroblasts, proteases and protease inhibitors) discussed in earlier sections. Relevant texts are on line 561-586.

(4) Consider adding a glossary of key terms (e.g., matrikines, efferocytosis) to aid accessibility.

A glossary explaining selected terms that may be confusing to the general readership is now added as Appendix 1 (line 977).

Reviewer #3 (Recommendations for the authors):

The discussion of fibrosis as a significant consequence of inflammatory activity is currently limited to skin keloids and bleomycin-induced lung fibrosis. Considering the substantial clinical relevance, it would be beneficial to include a mention of the various forms of liver fibrosis resulting from chronic inflammation.

Liver cirrhosis is now mentioned as further examples of stiffening tissues on line 428, 436-439.

While the manuscript is generally well-written, there are several minor language issues that could be easily addressed by a native speaker during revisions. Some examples are listed below:

We thank the reviewer for these very helpful suggestions. They are adopted with the relevant line number in the revised manuscript indicated below. In addition, the manuscript is proofread again, with other grammatical mistakes corrected throughout the text.

(1) Line 40: ... proliferative pathogen, can be timely eliminated.

line 40

(2) Line 79: It may be worthwhile pointing out that while Claudin 5 expression is highest in the BBB, it is also relevant in the BRB and expressed at lower levels in peripheral ECs. Similarly, ZO-1 is widely found to be expressed in peripheral endothelial cells.

Thanks for indicating this caution, it is now mentioned on line 79-82.

(3) Line 82: affects leukocyte traffic and...

line 84

(4) Line 125: ..., both neutrophil and lymphocyte extravasation were reduced by ~60%

line 125-126

1. Line 128: The term "paracellular endothelial junction" is odd, as junctions are per se paracellular, i.e., between cells.

line 129

(6) Line 147: ... VE-Cadherin, in which the FRET signal vanishes.

line 148

(7) Line 186: "activation by direct leukocyte pressing" might be rephrased to be clearer, e.g. "it might as well be activated by mechanical force exerted by leukocytes like it is the case for Piezo-1."

line 185-186

(8) Line 216: The phrasing "knockout analogy" is somewhat unfortunate. I would suggest "...a4 ko mice consequently largely lack a5 low expression regions and the resulting

| *reduction in leukocyte extravasation confirms the facilitating role of the low $\alpha 5$ expression regions."*

line 217-218

| *(9) Line 219: ...how the low expression regions form / are formed in the first place... The term construction implies active planning.*

line 220

| *(10) Line 278: ... thrombocytopenic mice ...*

line 279

| *(11) Line 294: ... use platelets as a drug delivery vehicle ...*

line 295

| *(12) Line 304: instead of "could have changed", use "might change"*

line 315

| *(13) Line 320: at the level of the monocyte*

line 336-337

| *(14) Line 324: ... consistent with ...*

line 340

| *(15) Line 335: ... progenitors*

line 351

| *(16) Line 432: ... a considerable number of apoptotic neutrophils has (been) accumulated*

line 480

| *(17) Line 442: ..., which promote killing responses, cross activate other leukocytes ..., or reduce tissue availability...*

line 490-491

| *(18) Line 453: ...This macrophage is responsive to BMP...*

This sentence is now rephrased on line 500-501.

| *(19) Line 454: ...involved in forming S1 macrophages.*

line 502

| *(20) Line 476: ...numerous pathologies...*

Points (20-22) concerns Section 11, which is now restructured (line 523-586).

1. Line 492: ...macrophages acquiring phenotypes specific to their residence tissue.

(22) Line 498: ...either - the tissue macrophage is of heterogeneous nature... or - tissue macrophages are of heterogeneous nature...

<https://doi.org/10.7554/eLife.108284.2.sa0>